



Is convexity efficiently priced? Evidence from international swap markets

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ABSTRACT

While it is widely claimed in the literature that convexity is correctly priced, we find evidence in four major swap markets that this is the case only on average and that extended periods occur when convexity-based trading strategies offer economically very significant exceptional returns. These abnormal returns can be reaped with fully no-peek-ahead strategies and after accounting for transaction costs. We find a strong link between the periods of highest profitability and conditions of reduced market liquidity. This suggests that, as observed in recent liquidity studies on US Treasuries, temporary deviations from market efficiency at the long end of the swap curve occur when pseudo-arbitrageurs do not have sufficient capital to correct the mispricings.

1. Introduction

Trying to make money from being systematically long convexity in the largest international swap markets is economically unattractive. We look at dollar-, euro-, sterling- and yen-denominated long-dated swaps and we find that, in all currencies, the strategy of systematically buying a long-convexity portfolio hedged against changes in the first few principal components is never economically attractive. The Sharpe ratios are either negative or, when positive, not statistically different from zero. More significantly, even a moderately risk-averse investor would not pay a manager engaging in this *unconditional* long-convexity strategy a positive fee — actually, the same risk-averse investor would often require a fee of several percentage points per year to be willing to accept the strategy. The picture changes radically if we look at convexity as an option that should be opportunistically bought or sold, with an associated self-financing dynamic trading strategy based on the slope of the yield curve, its curvature and the volatility of the underlying swap rates. In all the currencies analysed, using this information we find such a high degree of predictability for changes in the value of convexity that even a considerably risk averse investor would now be ready to pay a manager very high fees (of many percentage points per annum) to enjoy the benefits of this *conditional* convexity-based strategy.

We also look at a related but, in important respects, different conditional strategy, based on exploiting the mean-reversion of the swap curve curvature. With one exception discussed at length in Section 4, the returns from this strategy are also always positive, the Sharpe ratios sometimes very high and the management fees large, positive, and often larger than the fees payable to a convexity-based manager. Most of the times curvature-based strategies therefore seem to work as well as, if not better than, convexity-based strategies. We show, however, that they can fail badly in periods of financial turmoil, and that this can give rise to a large loss of utility. We explain in Section 5 that this happens because, lacking information about conditional volatility, these curvature-based strategies fail to recognize when swap curve ‘should’ display very high curvature.

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In this study we focus on the *economic* significance of the strategies we propose. Since we will argue that over extended periods of time investors in the most liquid derivatives markets in the world ‘leave money on the table’,¹ we have to show that this forgotten money can actually be collected. We therefore take the following steps to ensure the implementability of both (convexity- and curvature-based) strategies. First, both strategies strictly make use only of information available at trade time. Second, we take the perspective of an investor in a hedge fund, or of the proprietary desk of an investment bank, and we recognize that these investors will have to put up either regulatory market risk capital, or initial and variation margin, or both. Next, we also account for transaction costs — although we find that these do not qualitatively change our conclusions because their impact is small. Finally, we use daily prices of individual swaps for our analysis, while most of the convexity studies reported in the literature make use of the fitted prices of ‘virtual’ discount Treasury bonds. This is important because we are looking at a second-order effect and pricing errors introduced by fitting can be as large as, or bigger than, the effect we are exploring.

Key to understanding the origin of the profitability of the convexity-based strategy is the interpretation of the curvature of the swap curve as giving rise to an ‘option payoff’. Indeed, we show in Section 2 that a duration-immunized long-convexity strategy is exactly equivalent to a delta-hedged long-gamma (straddle) position in options, with a close correspondence between the theta (first derivative with respect to time) of the straddle and the time decay of the long-convexity position. As in the option case, in a diffusive setting at equilibrium the ‘premium paid’ by an investor who engages in a convexity strategy is recovered by continuously rebalancing the swap positions so as to remain neutral with respect to the generalized durations.² And, as again in the case of ‘gamma-trading’ with options, an investor will choose to be long or short convexity (after duration hedging) depending on whether she expects the realized volatility of the swap portfolio to be higher or lower, respectively, than the ‘implied volatility’. We show that in the swap market case the implied volatility is related the shape of the swap curve — in particular, to its steepness and its curvature.

The practical investment implications of our findings are self-evident. From the theoretical perspective, our results indicate that the long end of the swap curve is not always efficiently priced, and that there is an economically-exploitable degree of predictability in the changes in the swap curve curvature. More precisely, we show that *on average* the swap curve is not far from having a fair shape, but at any given point in time it can display either too much or too little curvature — where ‘too much’ or ‘too little’ are to be understood in the sense of allowing very-low-risk profitable strategies that would prove attractive even to a very risk-averse investor.

Since we propose these strategies for highly-developed and liquid markets, why do we expect them to be profitable? We find that the curvature of the swap curve is strongly variable, with a high reversion speed. Both our strategies are based on exploiting the movements of the curvature away from, and back towards, where ‘it should be’ (given different signals of the correct value). This interpretation of why exploiting the mean-reversion of the curvature can be profitable is similar (for a higher mode of curve deformation³) to the rationale behind the Cieslak and Povala (2015) return-predicting factor (that is established for the yield curve level). The similarity arises because part of the profitability from the Cieslak–Povala factor comes from the fast mean-reversion of their ‘cycles’, which can also be interpreted as deviations of the yield level from ‘where it should be’ (given, in their case, long-term inflation). We discuss this aspect in Section 5.3, where we show that the correlation between the profitability from the Cieslak–Povala and our strategies is highly significant.⁴

However, the origin of the profitability of the curvature- and convexity-based strategies we propose is even more closely aligned with the financial mechanism uncovered by Hu et al. (2012), who argue that mispricings in the US Treasury market occur in periods of turmoil, when the actions of pseudo-arbitrageurs are hampered by poor liquidity. In the case analysed by Hu, Pan and Wang, local deviations of yields of bonds of similar maturity are not arbitrated away during periods of illiquidity because arbitrage capital is not forthcoming. (Very aptly their paper is titled *Noise as Information for Illiquidity*.) We find that the curvature of the swap curve strays from its equilibrium level in the same periods of poor liquidity, offering trading opportunities that can be exploited when better trading conditions are re-established. We present a detailed analysis of this in Section 5, but, to motivate the reader, we conclude this introduction by showing in Fig. 1 the striking correspondence between the profits made by our curvature strategy in the US dollar swap market and one of the liquidity measures we discuss in detail in Section 5.

Our work contributes to the literature on yield curve convexity in several respects. First of all, the profitability of convexity has been explored in the literature without explicit reference to liquidity. So, for instance, Khan and Lochoff (1990) carry out a return attribution analysis by examining the impact on Treasury excess returns of several factors, one of which is convexity. They do not find the convexity factor to generate significant excess returns. Shyy and Lieu (1994) extend the study by Kahn and Lochoff by looking at the convexity effect on returns in the US Treasury bond market in the 1983–1987 period. They do so by sorting bonds into 13 classes with similar duration. Then, within each duration cohort they divide bonds into high-or-low convexity groups. Finally, they test by regressions the hypothesis that there should be a positive relationship between convexity and excess returns by comparing the high- and low-convexity groups. Again, they find a null result, from which they conclude that markets are efficient and any

¹ At the end of June 2018, the notional outstanding of interest-rate swaps in US dollars, British pounds, Euros and Yens were \$180, \$125, \$52 and \$50 trillion (notionals converted to US dollars at the end-of-June exchange rate) – source (BIS, 2018). Even if in the case of swaps the notional outstanding is a poor reflection of market risk, it is related to the volume traded, and hence to liquidity. For comparison, the outstanding notionals for the Treasury markets for the United States, United Kingdom, Germany and Japan were \$19, \$2.9, \$1.8 and \$9.4 trillion, again after converting to US dollars.

² By ‘generalized durations’ we refer to the sensitivities of swap prices to first-order changes in the underlying factors, whatever these may be.

³ We define precisely what we mean by ‘modes of deformation’ in Section 3: for the moment they can be thought of as similar to the first three principal components, i.e., generalized ‘level’, ‘slope’ and ‘curvature’.

⁴ In the Cieslak–Povala case, the equilibrium level of yields is obtained by regressing yields against a slow-moving average of inflation. The cycles are then the residuals from these regressions, and indicate deviations from the equilibrium level.

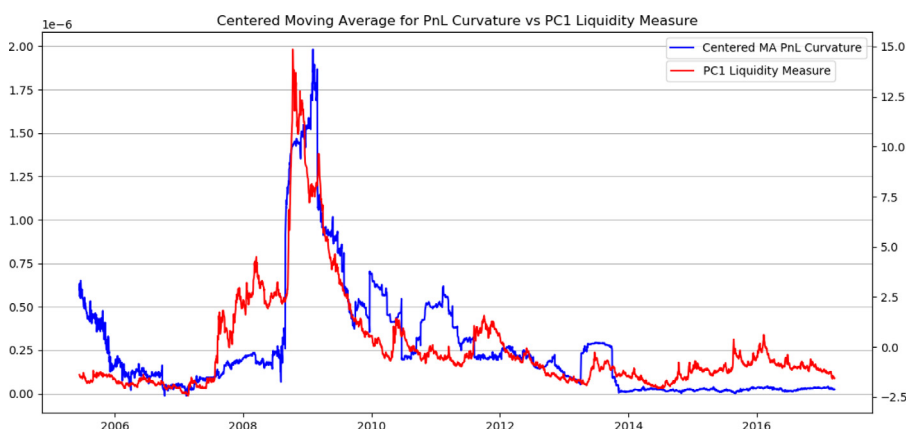


Fig. 1. The 60-day centred moving average of the profits made by the curvature strategy in the US dollar swap market (left-hand y axis) and the PC-based liquidity measure in Rebonato and Sherwin (2018) (right-hand y axis).

gains from convexity are ‘priced in’ by the market. Ben Dor et al. (2006) come to a similar conclusion when they study whether an immunized strategy only exposed to convexity can yield excess returns in the U.S. Treasury market in the period 1989–2006. Kang and Chen (2002) look at the government debt market in several countries and find that “the theta effect is [...] much more important than the convexity effect, even for a one-day period and for a yield change equivalent to two standard deviations”. Finally, Fabozzi (1999) looks at convexity from the perspective of market efficiency and concludes in very general terms that in an efficient market convexity is priced according to current yield volatility expectations. What all these studies have in common is that they analyse what we call the ‘unconditional’ strategy of being long convexity (after duration hedging) all the time and none of them considers any dynamic strategy. As we show in Section 4, we concur with their conclusions regarding these unconditional strategies, but we find the most interesting results in the *conditional strategies* that are the main focus of this study.

In an early study of the efficiency of the international swap markets, Gupta and Subrahmanyam (1998) looked at the differences between swaps and future contracts in dollar, sterling, Deutsche Marks and yen, and found that in their early days “swaps were being priced off the futures curve” (i.e. by ignoring the convexity adjustment) and that, after this initial period, “the market swap rates drifted below the rates implied by futures prices”. By using a number of term structure models (Vasicek, 1977; Cox et al., 1985; Hull and White, 1990), and Black and Karasinski (1991), they argued that the convexity bias was linked to the theoretical value of convexity, “with a gradual elimination of that mispricing by incorporation of a convexity adjustment in swap pricing”. They therefore concluded that, after the ‘convexity bias’ was taken into account, convexity was correctly priced. Despite apparent similarities, there are key differences between their work and ours. First, Gupta and Subrahmanyam and this study look at efficient pricing of swap convexity from two different angles: they ask the question whether, for a given volatility, swaps are correctly priced *relative to futures*. We explore whether one long-dated equilibrium swap rate is correctly priced *relative to neighbouring-maturity swap rates*. Therefore we investigate whether, again for a given volatility, the shape of the long end of the swap curve is such as to preclude abnormal returns. Second, our data set starts after the end of the data used by Gupta and Subrahmanyam, and should therefore reflect fully ‘mature’ swap markets. Any inefficiency we find cannot therefore be attributed to the swap market being underdeveloped. Finally, we note that at the time of their study, 30-year swaps, which are essential to bring to light a significant convexity effect, were not actively traded.

To our knowledge, the main study on conditional trading strategies based on convexity is in Rebonato and Putiatyn (2018). The main differences between their and our work are as follows. First, they look at Treasury bonds (rather than swaps) and use virtual discount bond prices obtained by best fitting to market prices from the well-known Fed data-set (Gurkaynak et al., 2007). Pricing errors, while generally small, can however be of the same order of magnitude as the second-order effect we are focusing on in this study. We do not have this problem because we study the swap market, with close-of-business individual prices (equilibrium swap rates and discount curves). Second, in their analysis Rebonato and Putiatyn only measure performance using the Sharpe ratio (which we show is very poorly suited to the task given the strong asymmetry of returns) and they do so by employing all-data (look-ahead) information. As numerous studies in fixed-income return-predicting factors testify (see, e.g., Dai et al. (2004) and Kessler and Scherer (2009)), making use of all-sample data can substantially flatter all performance measures. In contrast, we only use strictly backward-looking information that could be employed in real trading conditions and we account for transaction costs. By doing so, we check that the abnormal returns we identify can indeed be reaped in practice. Another source of difference is that Rebonato and Putiatyn do not consider the economic significance of their results. By employing two different non-gameable performance metrics, we assess whether the returns are economically, and not just statistically, significant. Finally, we take an international perspective and, by doing so, we establish some broad regularities at least among developed government debt markets.

The paper is organized as follows: in Section 2 we present a qualitative discussion of the strong similarities between convexity and long-gamma option strategies. These qualitative similarities with option strategies are quantitatively developed in Section 3, where we derive the trading signals build the associated portfolios. Here we also introduce the performance metrics used to assess

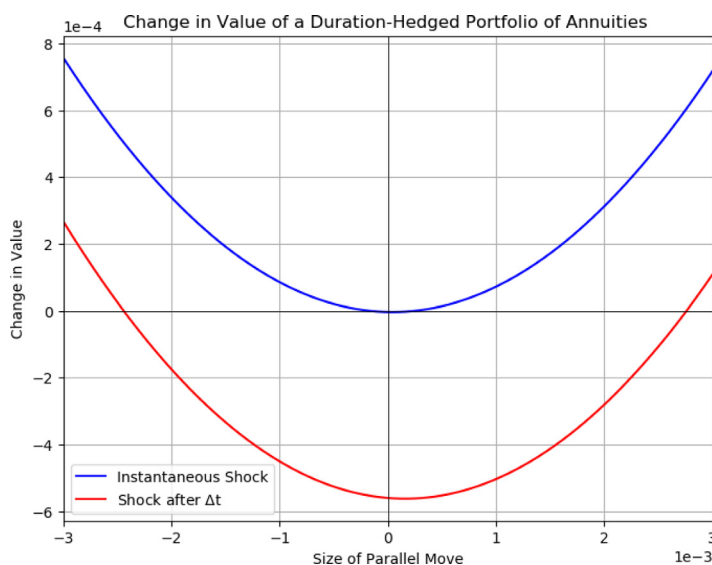


Fig. 2. The changes in value of a duration-hedged portfolios of annuities for instantaneous parallel moves of the yield curve of the magnitude reported on the x axis (upper curve), and for the same moves after a time $\Delta t = 0.1$ (lower curve).

the economic value of the various strategies. Our empirical results are presented Section 4. In Section 5 we discuss their implications and their robustness; we relate periods of high profitability for both strategies with periods of poor market liquidity; we explain why and when the curvature and convexity strategies perform differently; and we discuss the effects of transaction costs. The conclusions and our suggestions for future work are found in Section 6.

2. Convexity as optionality

Before delving in the details of the various strategies it is important to understand their financial underpinnings. The best way to do so is by exploring the close similarities between long-gamma option trading and long-convexity swap strategies. In this section we keep the discussion at a qualitative level, and present a quantitative treatment in Section 3.2.1.

The change in value of a swap for a change in rates has a term linear in the swap rate, and a convex term arising from the non-linear dependence of the annuity on moves in the swap curve. See Eq. (1). Since the linear term can be exactly hedged with linear positions in the swap rate, we focus in what follows on the non-linear term, related to the annuity. To motivate the discussion, Fig. 2 therefore shows the change in value of a duration-hedged portfolio of annuities (i) for instantaneous parallel moves in yields of magnitudes shown on the x axis (upper curve) and (ii) for the same yield moves after a finite time Δt (lower curve). This profile is qualitatively identical to the profit-and-loss profile from a delta-hedged long option position. In particular, we note that:

1. if time elapses and no move in yields occurs, then the annuity portfolio loses in time decay — this is exactly mirrored in the loss in theta value over a finite time interval for a delta-hedged option whose underlying does not move;
2. there are two points in Fig. 2 where the loss in time value is exactly compensated by the magnitude of the parallel move in yields — these two points are, in option language, the ‘break-even points’, where the theta decay is compensated by profit made on re-hedging;
3. as in the case of a delta-hedged long option position, also in the case of the duration-neutral portfolio of annuities whether the holder makes or loses money depends on the magnitude of the move in the swap rates.

The correspondence is strengthened by observing that in the case of delta-neutral, long-gamma positions the loss in time value (theta) is compensated by buying the stock when the stock price falls and selling when it rises. Similarly, a duration-neutralized portfolio of swaps with positive convexity will increase in duration as yields fall (and prices rise) and will have a shorter duration as yields rise (and prices fall).⁵ The investor in this long-convexity swap portfolio will therefore systematically make money from re-establishing a duration-neutral position. Both in the swap portfolio and in the option case, the more frequently the underlying (the stock price or the level of the swap curve) performs these up-and-down movements, the greater the profit made by re-hedging (buying low and selling high).

Establishing a long-gamma option position costs money. What is the equivalent of this paid-out premium in the case of the long-convexity swap portfolio? Suppose that we establish a long-convexity position by entering long receiver swap positions in

⁵ Unless otherwise stated, we always assume the case of a receiver swap, for which the intuition is aided by the similarity with a long position in a coupon bond.

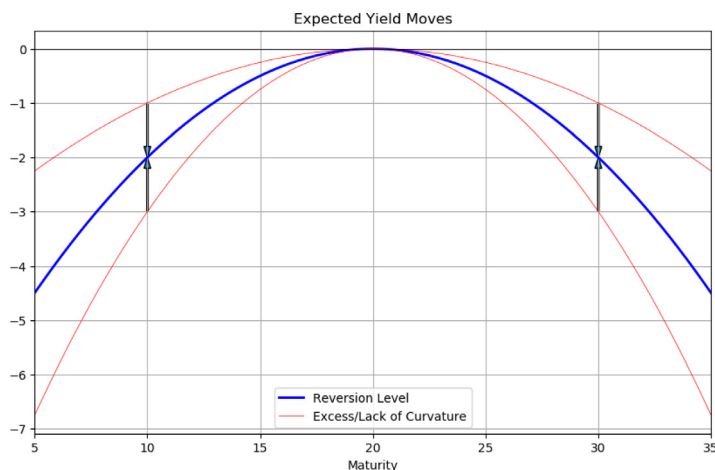


Fig. 3. Expected yield moves when the curvature reverts to its reversion level (thick curve) from above or from below. The swap maturity (in years) is reported on the x axis. When the curvature is more negative than its reversion level (bottom curve), a curvature-based strategy suggests a short position in the 30- and 10-year swaps and a long position in the 20-year swap. As the curvature moves to its reversion level, a profit is made from the 30- and 10-year swap rate moving up (as shown by the bottom black arrows). The reverse occurs when the curvature is less negative than its reversion level (top curve).

maturities T_1 and T_3 , and a short receiver swap position in maturity T_2 , with $T_1 < T_2 < T_3$ and $T_1 \gg 1$ – so that we are looking at long maturities. Let K_1 , K_2 , and K_3 be the fixed rates for the three swaps and consider the common case of an upward-sloping swap curve.⁶ Suppose that we choose the maturities in such a manner that the weights for the three swaps are $w_1 = 1$, $w_2 = -2$, and $w_3 = 1$. Now, as Brown and Schaefer (2000) have shown, because of convexity and the high persistence of rates, for sufficiently long maturities yield curves should always be declining *and should do so with a negative second derivative*.⁷ In this stylized example, over a time interval Δt the net accrued interest received ($K_1 + K_3$) by the investor *will therefore always be smaller than the interest paid* ($2K_2$). This deterministic amount is purely a function of the shape of the swap curve, and is what traders call ‘negative carry’. For the swap portfolio, this negative carry is the equivalent of the option premium paid over each time interval Δt . The more negative the curvature of the swap curve, the greater the ‘premium paid’ and the higher the realized volatility has to be for the portfolio holder to be compensated for the time decay. At equilibrium there must therefore be a link between the volatility of long-dated swap rates and the curvature of the swap curve for long maturity. The convexity trading strategy we propose exploits this link by identifying states of the world when the swap curve displays too much or too little curvature (given the expected volatility over the next re-balancing step). Gamma trading rests on the interplay of realized and implied volatility; in the case of the swap portfolio the role of the implied volatility is therefore played by the curvature of the swap curve.

In addition, if the curvature is on average correct, but not at all times ‘what it should be’, moving away from and then reverting to its equilibrium level, then a strategy based on the mean-reversion of the curvature may also be profitable. Note that this second strategy is quite distinct, as it does not require a view on the future volatility. The potential source of profitability of the strategy based on the reversion level of the curvature can be understood from Fig. 3, which refers to a hypothetical duration-neutralized portfolio of 10-, 20- and 30-year swaps. Given mean reversion, the curvature will tend to revert to its reversion shape from below when too negative and from above when not negative enough. In the first case, the investor will be short the 10- and 30-year swaps in the portfolio and will benefit as these rates rise to revert to the long-term level of curvature. The opposite will occur if the curvature is too shallow. (Since we ensure that we are immunized against changes in level, without loss of generality we can assume that the mid-maturity swap rate remains unchanged.) We study both these strategies in what follows. We stress again that the goal of this section has merely been to build the intuition and provide a motivation for the two strategies. As we said, the precise assumptions and the quantitative treatment are provided in Section 3.2.1.

3. Construction of the strategies

3.1. Building the duration-neutral long-convexity portfolio

The value at time t of a receiver swap with maturity T_i and N periods to expiry is given by

$$S_t^{T_i} = [K - SR_t^{T_i}] Ann_t^{T_i} \quad (1)$$

⁶ The condition is not necessary and only introduced to simplify the argument.

⁷ The sufficient condition for this to occur is that the volatility of the long term zero coupon yield should be “sufficiently high”; in practice this means that, at least over a significant maturity range, it should not decline more quickly than T^{-2} .

where $SR_t^{T_i}$ is the time- t equilibrium rate of the swap, K its fixed rate and $Ann_t^{T_i}$ the time- t value of its annuity. The annuity, $Ann_t^{T_{N_i}}$, is given by

$$Ann_t^{T_{N_i}} = \sum_{i=1, N} P_t^{T_i} \tau_i \quad (2)$$

where $P_t^{T_i}$ is the time- t discount factor for maturity T_i and τ_i denotes the length of the i th accrual period (approximately half a year in our case).

Our goal is to create a long-convexity portfolio first-order immunized against changes in the swap curve. Consider first the case when we want to duration-hedge against changes in the ‘level’ and the ‘slope’.⁸ Statistically, the most efficient way to do so would be to hedge against changes in the first and second principal components. With the data at our disposal, however, calculating the principal components would entail either using whole-sample information not available at the time of trading or throwing away a substantial fraction of the data for the estimation period. We want to use all the data at our disposal at a given point in time, and to have a no-peek-ahead strategy. We therefore define three or four fixed linear combination of swap rates, x_t^i , judiciously chosen as to mimic the principal components of a Toeplitz matrix (see below). Because of their similarity to principal components, we refer to these linear combinations as ‘modes of deformation’. We assume that changes in these modes of deformation, given in matrix form by

$$\Delta x_t = A \Delta S R_t, \quad (3)$$

explain changes in the whole swap curve for the universe of swaps in our portfolio. In [Appendix A](#) we specify the form of the matrix A for three and four factors and derive that the weights w are given by

$$w = C^{-1}y \quad (4)$$

with the matrix C and the vector y also derived in [Appendix A](#).

Effective hedging is predicated on a good choice of the A matrix. We were inspired in building our no-peek-ahead A matrix by the observation that the elements of the covariance matrix of swap-rate changes decay approximately exponentially with ‘distance’, $|T_i - T_j|$, (see in this respect, the extensive discussions in [Schoenmakers and Coffey \(2003\)](#) and [Rebonato \(2002\)](#)). When this is the case, the covariance matrix can be well described by a Toeplitz matrix, whose eigenvectors are approximately equal to the columns of the discrete cosine transform matrix. (See, e.g., [Ahmed et al. \(1974\)](#) and [Strang \(2003\)](#)). We also note that [Crump and Gospodinov \(2019\)](#) show that when the short rate is very persistent the loadings on the principal components for forward rates are perturbations of sine functions of increasing frequency. Therefore we translate these findings in our matrix A by imposing that all loadings should be orthogonal to each other, and that of each additional factors should have one more crossing of the x axis, so as to mimic the behaviour of harmonic functions of increasing frequency.

Appealing as they are, these justifications for the values of the elements of A are purely theoretical in nature. Do they work in practice? To answer this question, first we note that [Diebold and Rudebusch \(2013\)](#) show in the closely-related case of US Treasury yields that the time series of our variables x_t , $i = 1, 2, 3$ very closely mirror the time series of the first three principal components.⁹ We then check that this is indeed the case for the case at hand by comparing the all-sample first three or four principal components (to which we have no access in building our strategies) and the no-peek-ahead factors x_t^i that we do use for hedging.¹⁰ The results from one such typical set of tests (for USD) are shown in [Fig. 4](#), which displays the time series of the all-sample first four principal components and of the factors x_t^i , $i = 1, 2, 3, 4$ obtained with the 4×4 matrix A in the case of USD. The correlation between the first four principal components (to which we do not have access for our strategies because they are estimated in-sample) and the out-of-sample factors x_t^i are 99.99%, 99.28%, 98.57% and 89.19%. Even closer agreement was obtained with a 3×3 matrix A (not shown).

From these comparisons we can therefore derive high confidence that the chosen factors will first-order immunize very effectively against the changes in the swap rates, and that our portfolio will indeed be almost exclusively exposed to the second-order (convexity) effects that we want to capture.

A portfolio of swaps with weights given by [Eq. \(4\)](#) will be long convexity and immunized against changes in the first two or three x_t factors. We call this the *reference convexity portfolio*. We note that the solution w is only determined to within an arbitrary non-zero multiplicative factor. The reference convexity portfolio corresponds to the choice of $w_N = 1$ (where N is the maturity of the longest-maturity swap). With this choice, however, at any given time the convexity of the reference portfolio will be different. We therefore scale the reference portfolio by its net convexity to obtain a *normalized reference convexity portfolio* with constant convexity at each point in time. For brevity, we sometimes more concisely refer to this as the ‘unconditional portfolio’.

Finally, we note that the portfolio associated with the curvature-based strategy described in [Section 3.2](#) must be exposed to the third mode of deformation, the ‘curvature’. Therefore it should only be hedged against the first and second factors, x_t^1 and x_t^2 , if

⁸ It has been known since ([Litterman and Scheinkman, 1991](#)) that a very high fraction (up to 98%) of yield curve variability can be explained by the first three principal components, which have been interpreted as ‘level’, ‘slope’ and ‘curvature’ because of the sign and magnitudes of the first three eigenvectors. This view has been recently challenged by [Crump and Gospodinov \(2019\)](#), who argue that the effective dimension of the data-generating process for interest rates could be as low as 1. Even in this case, [Crump and Gospodinov \(2019\)](#) show that the resulting eigenvectors still have the familiar level/slope/curvature shape.

⁹ See their [Fig. 1.4](#) and the discussion in their [Section 1.4](#).

¹⁰ For ease of visual comparison we have made the already orthogonal vectors of the matrix A also normalized. This has no effect on the hedging results.

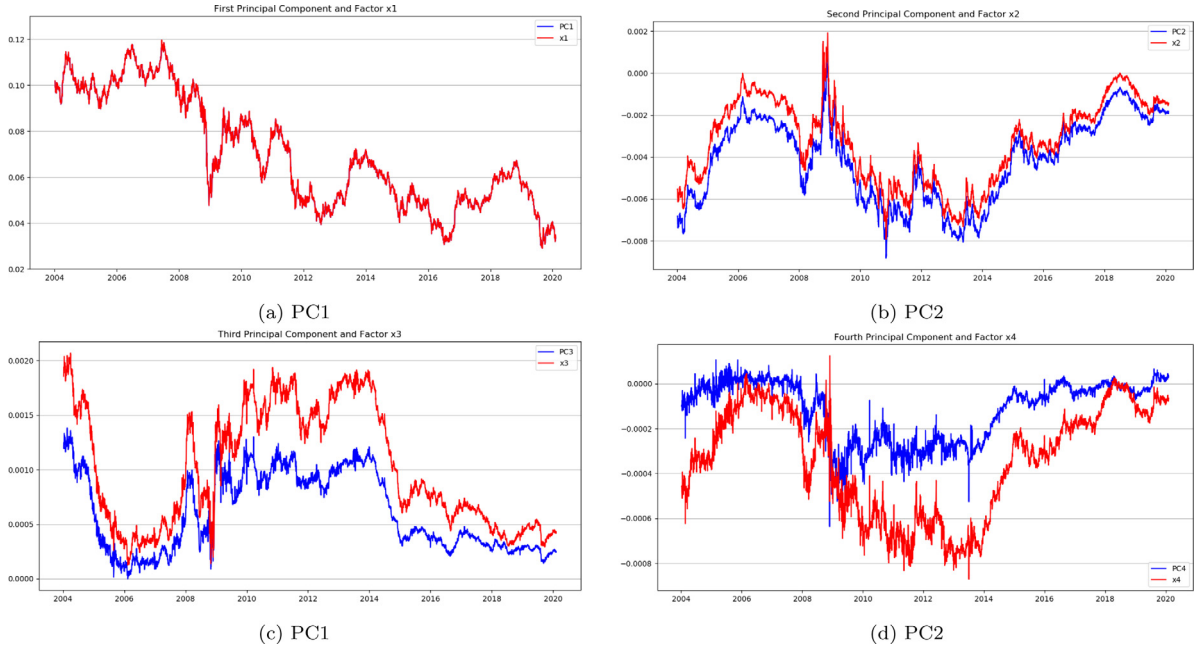


Fig. 4. In the upper row, the first two all-sample principal components (labelled ‘PC1’ and ‘PC2’) and the factors x_t^1 and x_t^2 in Eq. (3) in the case of USD. Note that in the left-hand panel the two curves are so close as to be indistinguishable. In the lower row, the third and fourth all-sample principal components (labelled ‘PC3’ and ‘PC4’) and the factors x_t^3 and x_t^4 in Eq. (3) in the case of USD.

it is to capture value from *first-order* changes in the third. The convexity strategy, on the hand, is designed to be affected only by *second-order* changes, and ideally one would therefore like to immunize the associated portfolio against first-order moves in all the underlying factors. However, to immunize against the first n factors one must use $n + 1$ swaps, and the resulting portfolio therefore remains exposed to the $(n + 1)$ th factor (fortunately, the one of smallest volatility, if the factors x_t^i have been properly chosen). As a consequence, *total* duration immunization is not achievable, and there is no ‘correct’ number of principal components against which the portfolio should be immunized. Rather, there is a trade-off between effectiveness of the duration immunization, out-of-sample robustness of higher and higher principal components, and ease of interpretability of the resulting long-convexity portfolio as more and more swaps are added.

Empirically we find that duration immunization against changes in x_t^1 and x_t^2 provides the most robust and the most easily interpretable results. Therefore we report in the body of the paper the findings pertaining to this choice (two-factor, three-swap convexity portfolios). However, in the online technical appendix (worksheet ‘Performance’) we also report the (qualitatively similar) results obtained for the convexity strategy after duration immunizing against changes in x_t^3 as well (with three-factor, four-swap convexity portfolios).

3.2. Building the signals

We start our analysis by first assessing the profitability of the benchmark (unconditional) strategy of always being long the normalized duration-neutralized portfolio. We then consider the improvement in economic performance over this unconditional strategy obtained by scaling at each point in time its notional by the strength of different state-dependent signals.

The signals depend in the case of the convexity strategy on the expected change in value of the reference convexity portfolio (obtained using Eq. (12)); and in the case of the curvature strategy on the distance of the curvature from its reversion level. In the following we therefore explain how these two signals are built: Section 3.2.1 deals with the convexity signal, and Section 3.2.2 with the curvature signal.

3.2.1. Construction of the convexity signal

Recall that the value of a *receiver* swap is given by

$$S_t^{T_i} = [K - SR_t^{T_i}] Ann_t^{T_i} \quad (5)$$

Let us make the assumption that the swap rate at time t is a sufficient statistic to describe the annuity,¹¹ by setting:

$$Ann_t^{T_{N_i}} = \sum_{i=1, \frac{T_N}{\tau}}^{T_{N_i}} P_t^{T_i} \tau \quad (6)$$

¹¹ This is the assumption underlying the determination of the value of cash-settled (as opposed to physically-settled) swaps in the market.

and

$$P_t^{T_i} \approx \sum_{k=1, \frac{T_i}{\tau}} \frac{1}{(1 + SR_t^{T_i} \tau)^k} . \quad (7)$$

Then we can write $Ann_t^{T_{N_i}} = Ann_t^{T_{N_i}}(SR_t^{T_{N_i}})$. By Ito's lemma, the time- t change in value, dS_t^T , of a new equilibrium swap of generic maturity T that settles immediately after inception is then given by

$$dS_t^T = \frac{\partial S_t^T}{\partial t} dt + \frac{\partial S_t^T}{\partial SR_t^T} dSR_t^T + \frac{1}{2} \frac{\partial^2 S_t^T}{\partial dSR_t^T{}^2} \sigma_{SR}^2 dt . \quad (8)$$

We show in [Appendix B](#) that this gives for the expected value of the change in value of the swap¹²:

$$E_t[dS_t^T] = [K - f_t^\tau]dt - Ann_t^T(SR_t^T) dSR_t^T - \frac{\partial Ann_t^T}{\partial SR_t^T} \sigma_{SR}^2 dt \quad (9)$$

where f_t^τ is the time- t value of the τ -period LIBOR rate that resets at inception.

For the term dSR_t^T we make the following assumptions:

1. given the long maturities, T , of interest in convexity studies (10 to 30 years) the expectation contribution to the change in the swap rates is negligible; and
2. the maturity-dependent risk premium remains the same over the time period dt .

Under these joint assumptions,¹³ the term dSR_t^T can be written as

$$dSR_t^T = SR_{t+dt}^{T-dt} - SR_t^T \approx SR_t^{T-dt} - SR_t^T \quad (10)$$

Multiplying and dividing by dt we therefore have

$$\frac{SR_t^{T-dt} - SR_t^T}{dt} dt \approx -\frac{dSR_t^T}{dT} dt \quad (11)$$

with the term $\frac{dSR_t^T}{dT}$ representing the time- t slope (derivative with respect to maturity) of the swap curve. The expectation of the change in value of the receiver swap is therefore given by

$$E_t[dS_t^T] = [K - f_t^\tau]dt + Ann_t^T(SR_t^T) \frac{dSR_t^T}{dT} dt - \frac{\partial Ann_t^T}{\partial SR_t^T} \sigma_{SR}^2 dt \quad (12)$$

Note that as swap rates increase the value of an annuity decreases, so the term $-\frac{\partial Ann_t^T}{\partial SR_t^T}$ is positive. For an upward sloping term structure and a receiver swap the term $\frac{dSR_t^T}{dT}$ is also positive, and so is the $K - f_t^\tau$ term. For a monotonically upward-sloping yield curve all the terms in Eq. (12) would therefore be positive for a receiver swap. We note in passing that, since this cannot be true without allowing arbitrage, this confirms from a different perspective that, as the aforementioned ([Brown and Schaefer, 2000](#)) point out, beyond some point 'long-term interest rates should (almost) always slope downwards'. Of course, when we consider a duration-neutralized portfolio with weights w , the weighted sums of the individual terms could be positive or negative.

As mentioned, the idea is to scale the unconditional convexity portfolio by the strength (and sign) of the convexity signal. This is done as follows. Given the expectation for the changes in value of n swaps of maturities $T_1, T_2, \dots, T_n$ (with $n = 3$ or 4) obtained using Eq. (12), we therefore create a convexity signal, Sig_t^{conv} , at time t_i given by

$$Sig_t^{conv} = w_i^T E_t[dS_t^T] \quad (13)$$

where w_i is a three- or four-dimensional vector determined as in Section 3, T denotes the transpose, and $E_t[dS_t^T]$ is also a three- or four-dimensional vector containing the expected changes in value of the swaps in the portfolio, obtained from Eq. (12). The intuition is that, if the expected value of the reference normalized long-convexity portfolio obtained in Section 3 is positive (negative), we want to be long (short) this portfolio. These long or short positions are then scaled by the magnitude of the signal — i.e., the expected portfolio return. (Of course, multiplying at each point in time a duration-neutral portfolio by a constant preserves the duration neutrality.)

Finally, we note that to calculate the expected portfolio return we need the swap-rate volatilities (see the last term in Eq. (12)). These were estimated using a fully-backward-looking moving window, with equally weighted returns. More precisely, we used for the estimation a window initially expanding up to the length of one hundred business days, and then sliding forward using the last hundred business days of data. One could, of course, use more sophisticated volatility estimators, but we checked that even a forward-looking, all-sample GARCH(1,1) estimation would offer little improvement in performance over the measure we used. In order to avoid 'model optimization' over an array of backward-looking volatility estimates, we have therefore employed the simplest no-peek-ahead methodology and kept the length of the window to a reasonable round number (100 business days).

¹² We note in passing that there is no missing $\frac{1}{2}$ term in the last term of Eq. (9), because the derivative $\frac{\partial^2 S_t^T}{\partial dSR_t^T{}^2}$ carries a multiplicative factor 2 — see Eq. (45).

¹³ In market parlance, these assumptions are referred to as the 'yields-don't-move' assumptions, and are valid if the slope of the yield curve is purely due to (constant) risk premia. In the US Treasury market, [Cieslak and Povala \(2016\)](#) show that the changes in expectations explain the variability of yields only at the short end of the maturity spectrum.

3.2.2. Construction of the curvature signal

We now move to explaining how we built the signal for our second, curvature-based strategy. In all the currencies under study our third factor, x_3 , displayed strong mean-reversion, with a reversion speed very similar to what obtained for the all-sample third principal component. As discussed in the Introduction and in Section 2, these fluctuations in curvature may indicate temporary, mean-reverting deviations from where the curvature ‘should be’.¹⁴ If this were the case, we could exploit these temporary deviations from efficiency by scaling the normalized reference convexity portfolio by the deviation of the curvature from its reversion level.

The problem with this appealing approach is that in order to estimate the reversion level of the curvature an investor would need full-sample data, not available at the time of trade. We therefore create two backward-looking estimates of the curvature reversion level, θ_i , the first by using an ever-expanding data window starting at the beginning of the data set, and the second by using a rolling fixed-length window, W . So, in the both cases the curvature signal, Sig_i^{curv} , at time i is given by

$$Sig_i^{curv} = curv_i - \theta_i \quad (14)$$

where $curv_i = x_3(t_i)$ is the curvature at time i . Then we have for the ever-expanding estimate of the reversion level

$$\theta_i = \frac{1}{i} \sum_{k=1,i} curv_k, \quad (15)$$

and for the expanding-window estimate

$$\theta_i = \frac{1}{W} \sum_{k=1,W} curv_{i-k+1}. \quad (16)$$

As discussed in Elliot and Timmermann (2016), if one wants to mimic the plausible behaviour of an investor faced with the task of deciding how best to use the finite data at her disposal, these are the two psychologically most appealing options: in the former case the investor is agnostic about the possibility of structural breaks or, in general, lack of time homogeneity, and therefore gives the same weight to all the information at her disposal; in the second case, the investor tries to account for the possibility of time-varying behaviour of the time series by giving more weight to the most recent information. In this second case we have therefore implicitly used the time- t weights, ω_i , given by $\omega_i = 0$ for $i \leq t - W$ and $\omega_i = 1$ otherwise.¹⁵ We have not explored other variations (such as using a fixed fraction of the available data, also mentioned in Elliot and Timmermann, 2016), that we find more difficult to justify from the perspective of what an actual investor would do.

3.2.3. Mean–variance optimality and robustness of the portfolio

A risk averse investor would not simply scale the notional of the normalized reference portfolio by the strength of the signal (linked to the expected return), but would also take the portfolio risk into account. It is well known that volatility is predictable and highly persistent. One can therefore approximate the risk/return trade-off by scaling the conditional strategy by the inverse of the conditional variance. As Moreira and Muir (2017) show, given the time- t expected return for a given strategy, $E_t[r(t + dt)]$, a mean–variance investor would then commit to the strategy an amount proportional to the ratio of the expected return to the time- t conditional variance. By so doing, the investor would maximize expected return for a fixed variance, by taking more (less) risk when the portfolio variance was low (high). We therefore calculated the performance metrics described in Section 3.3 also in the case of variance-normalized conditional portfolios.

In addition to this, we wanted to test the robustness of the signal to a possible misestimation of the exact size of the expected return. We therefore also examined the case when an investor believed in the sign information conveyed by the curvature or convexity signals, but were sceptical about the reliability of the information conveyed by their magnitudes. To reflect the behaviour of such an investor, the absolute value of the notional of the normalized reference portfolio was left unchanged, but its sign changed to match the sign of the signal.

Both variations on the base conditional strategies (scaling by variance, and using only the sign information in the signal) produced on average small improvements (especially in EUR) with respect to the basic signals in the Sharpe ratios and in the performance statistics discussed in Section 3.3. However, since no qualitative differences were brought about by these additional analyses, to keep the message simple and uncluttered we only report in what follows the (more conservative) results for the basic conditional strategies described in Sections 3.2.1 and 3.2.2. The full results for the two variations are reported in the technical appendix available online.¹⁶

3.3. Performance measures of the strategies

As we stated in the Introduction, our focus is on the economic, rather than statistical, significance of the strategies we analyse in this paper. Given the very pronounced skewness of the return distribution, the Sharpe ratio conveys poorly information about the risk/reward profile of the various strategies. We therefore also employ as performance measures (i) the non-gameable theta measure

¹⁴ Of course, this need not be the case: the curvature may be fair at all times, and its changes simply reflect changes in the level of the mean-reverting volatility. We test for this possibility in Section 5.

¹⁵ As discussed in Elliot and Timmermann (2016), there is an obvious trade-off between using more recent, and therefore possibly more relevant, data, and the increase in statistical uncertainty due to the smaller sample size. We address this issue in Section 4 by considering windows of different length.

¹⁶ See worksheet ‘Performance’ in the online appendix.

discussed in Goetzmann et al. (2007), and (ii) a similar measure based on the certainty equivalent of a power utility function that we introduce and discuss below.

For our (unfunded) application, the theta measure, Θ , by Goetzmann et al. is defined as

$$\Theta = \frac{1}{(1-\rho)\Delta t} \log \left(\sum_{t=1}^T \left[\frac{1+r_t}{1+r_t^f} \right]^{1-\rho} \right) \quad (17)$$

and can be interpreted as the annualized continuously compounded excess return certainty equivalent of the portfolio. That is, a risk-free portfolio earning $\exp[\log(1+r_t^f) + \Theta\Delta t]$ each period would have a measured performance of Θ . (In Eq. (17) r_t is the non-annualized Δt -period return from the strategy, r_t^f is the non-annualized Δt -period risk-less rate, ρ is a risk-aversion coefficient and Δt is the rebalancing period.)¹⁷ Since in our case we have an unfunded transaction, we have to define the strategy return. This we do after introducing the second performance measure we use.

To check the robustness of our findings about the economic significance of the various strategies, we also make use of an additional performance measure, obtained as follows. We choose a power utility function, characterized by a coefficient of relative risk aversion, γ . With this we calculate the expected utility from the strategy and the certainty equivalent using as initial wealth the capital defined below. So, we assume that the investor commits all of her wealth to the capital required for the strategy. This takes us back to the question of what the capital should be.

Since the strategy involves equilibrium swaps of zero value at inception, without margin or capital requirement any strategy with a positive return would be an arbitrage (would have an infinite expected performance measure). In reality, *some* capital will have to be set aside either for regulatory purposes (if the investor were, say, the trading desk of a regulated financial institution) or to meet initial and variation margin commitments. We therefore use as an estimate of this capital the Basel II market regulatory capital, X , which is approximately stipulated to be

$$X = VaR_{10}(99) \times M \quad (18)$$

where $VaR_{10}(99)$ is the 10-day, 99-percentile Value at Risk generated by the strategy and M is a multiplier floored at 3, and in practice set by regulators between 3.5 and 6. See, e.g., BIS (2019). We stress that this is a lower bound to the capital required: for instance, a regulated bank will have to pay regulatory capital (which currently includes additional charges from ‘stressed VaR’) and post collateral to a Central Clearing Counterparty.¹⁸ For the sake of simplicity, we approximately account for these joined capital requirements by setting the multiplier, M , equal to 10. We stress that, by using a high value for the multiplier, we reduce leverage, and therefore produce *conservative* results.

This estimated capital then constitutes the denominator required to calculate the return, r_t , in Eq. (17), and the initial wealth in the calculation of the expected utility (and hence of the certainty equivalent, CE). In our context, the certainty equivalent is the amount of cash an investor would be willing to exchange for the strategy in question. In the spirit of Thornton and Valente (2012), we therefore estimate the management fee, F , that a risk-averse investor with a relative risk aversion coefficient γ would be willing to pay to a manager who carried out the strategy under consideration as

$$F = \frac{CE(\gamma) - X}{X} . \quad (19)$$

For comparability with the Goetzmann et al.’s Θ measure, we calculated the certainty equivalent needed in Eq. (19) using a power utility function with coefficient of relative risk aversion, γ , equal to 3.

We tested the statistical significance of the various performance measures as follows. For each conditional strategy, we set as our null hypothesis, H_0 , the statement that the convexity and curvature signals contained no information. More precisely, recall first that, for each strategy, its notional is proportional to the relative signal. We then tested whether, *given the series of notionals produced by the strategy*, the returns obtained by the particular ordering of notionals established by our strategies were different from zero only by chance. In order to test this, we carried out 1000 random permutations (‘shuffles’) of each signal; determined from the shuffled signal the notional of each conditional strategy; recalculated the performance statistics; and obtained their simulated distribution. From this, we finally calculated the associated p values.¹⁹

Having defined the performance measures we are going to employ, we move to presenting the data and the results.

¹⁷ The coefficient ρ should be selected so as to make holding the benchmark optimal for an uninformed manager. Using a diversified portfolio of stocks as the benchmark, the Authors argue that ρ should be between 2 and 4. We have used 3 for our analysis.

¹⁸ The transactions in question give rise to market risk, but also counterparty credit risk. For a regulated financial institution, both sources of risk give rise to a capital charge.

¹⁹ We did not carry out a traditional resampling (bootstrapping) test, because the link between the signal and the volatility prediction poses non trivial problems of consistency between the realized history of simulated swap rates, the curvature of the simulated swap curve and the non-peek-ahead volatility. More precisely, if convexity were efficiently priced, the curvature of the bootstrapped swap curve, determined by the random resampling, would have to be related, via the simulated realized volatility of the swap rates, to the squared changes in the past bootstrapped swap rates. We are not aware of established methodologies to do so.

4. Empirical analysis

4.1. The data

We used for the study daily close values of equilibrium swap rates²⁰ for swaps of maturities of 5, 10, 15, 20 and 30 years for the USD, GBP, EUR and JPN swap markets.²¹ The data cover the same period for all swaps from 6-Jan-2004 to 6-Feb-2020 and were provided by the asset manager PIMCO from their proprietary data base. To our knowledge this is the only convexity study carried out with actual individual swap rates rather than synthetic yield curves.

For each time series the exact one-day changes in values for the swaps were obtained using the full discount curve built at time $t + 1$ with dual-curve construction (i.e., by using a curve for projecting rates and one for discounting – see, e.g., Henrad (2014)), so no approximations were used in computing the daily profits. Notionally, we assumed that every day we would close out the previous day's position and we would establish a new position in the required amounts using the new equilibrium swaps. Needless to say, in practice we would not follow such a procedure, which is only used to simplify the analysis.

One important feature must be mentioned. In the EUR time series of swap rate changes we observe very large moves of opposite sign over two consecutive days (from 2 December to 3 December 2008) for the 30-year and 20-year swap rates. These moves correspond to shocks of 10.7 and 9.7 standard deviations for the *ex-ante* distribution of daily rate changes, and to a 14.4 standard-deviation shock for the *ex-ante* two-day return distribution. These magnitudes are so large that they significantly affect (negatively) the performance measures for EUR. We have not identified in other currencies moves of similar magnitude on those dates, nor have we been able to identify on those particular dates events in the euro area that could justify such a violent reversal (the period was, however, very volatile). We have not been able to ascertain with certainty from independent sources if the recorded EUR moves are genuine. We have nonetheless decided *not* to windsize or otherwise correct our data because

- the two dates are in a period of very high volatility for the EUR;
- EUR swap rates of similar maturities display similar, albeit more muted, up-and-down movements;
- in convexity studies large moves are at the heart of the profitability (or otherwise) of the associated strategies – one must therefore be particularly cautious in culling those moves;
- when utility-function-based performance measures are used, keeping the exceptional moves gives a more conservative picture of the profitability.

We show in Table 1 the salient distributional features of the changes in the 10-, 15-, 20- and 30-year swap rates for the four currencies under study. For the sake of brevity, additional descriptive statistics for the swap rates are reported in the technical appendix available online. Here we simply make two observations. First, rates have fallen very significantly (and, as the technical appendix shows, with a high degree of correlation – see worksheet “Swap Rates”) in all currencies – careful hedging against level changes risk is therefore essential. Second, for all currencies the distributions of swap-rate changes display little or no skewness and moderate leptokurtosis. This should be compared with the very different results for these statistics for the strategy returns, discussed in Section 4.

4.2. Empirical results

The quantitative analysis of the results is greatly aided by examining first the cumulative returns from the curvature and convexity strategies for the four currencies, which are displayed in Fig. 5.²² More precisely, for each currency the figures show the cumulative profits reaped by engaging in the curvature-based strategy after immunizing against level and slope (curve labelled ‘Curvature’); and in the convexity-based strategy after immunizing against level and slope (curve labelled ‘Convexity’). The (qualitatively similar) results for the convexity-based strategy after immunizing against level, slope and curvature are reported in the online technical appendix. See worksheet ‘Cumulative Conv 3-fact’. Comparing the different currencies, some strong similarities and one striking difference are immediately apparent.

In GBP for both strategies there is an initial period of high and virtually constant profitability from Jan-04 to Sept-06. From Sep-06 to the start of the Lehman crisis (late September 2008) the cumulative profit is totally flat. After a period of (surprisingly limited) turbulence in late-2008/early-2009, the cumulative profitability climbs again, but at a slow rate until Jul-2013. After that the cumulative profit is flat again. In USD for both strategies we have a linear increase in the cumulative profits from Jan-04 to Sept-08. After the Lehman turbulence the profitability grows again, but at a diminishing rate, until Nov-2013. After that there is no further increase in the cumulative profits. In JPN for both strategies we also observe a period of increasing profitability in the first period (from Jan-4 to the Lehman crisis), but at a declining rate. After a short period of turbulence, the cumulative profitability then grows at a low but constant rate until the end of the sample. The picture is very different in EUR. After an initial period (Jan-04 to May-08) of increase in cumulative profit at a declining rate, we enter an extremely volatile period (up to March-09) – a period that includes the exceptionally large movements in December 2008 discussed in Section 4.1. The curvature strategy (that ‘knows

²⁰ An equilibrium swap rate is defined at the swap rate that gives equal value to the fixed and floating leg of a swap.

²¹ In what follows we describe the dollar, yen, sterling and euro swap markets as the USD, JPN, GBP and EUR markets, respectively.

²² We discuss in what follows the results for the *more conservative* case of an ever-expanding window for the estimation of the reversion level of the curvature. The sensitivity of the results to the type of window – ever-expanding or fixed length – and, in the latter case, to the window length is presented in Table 4 and discussed below.

Table 1

The daily average, standard deviation, skewness and excess kurtosis for changes in the 10-, 15-, 20- and 30-year swap rates (labelled ‘ Δ SR10’, ‘ Δ SR15’, ‘ Δ SR20’ and ‘ Δ SR30’, respectively), for the four currencies under study.

Statistic	Δ SR10	Δ SR15	Δ SR20	Δ SR30
USD				
Average	-7.5×10^{-6}	-8.4×10^{-6}	-8.8×10^{-6}	-8.9×10^{-6}
St Dev	0.0006	0.0006	0.0006	0.0006
Skewness	-0.0026	-0.0470	-0.0498	-0.0863
Kurtosis	3.3805	3.3010	3.6207	4.2438
GBP				
Average	-1.1×10^{-5}	-1.0×10^{-5}	-1.0×10^{-5}	-1.0×10^{-5}
St Dev	0.0004	0.0004	0.0004	0.0004
Skewness	-0.0359	-0.0645	-0.0793	-0.1120
Kurtosis	1.5963	1.6710	1.8059	1.7665
EUR				
Average	-1.1×10^{-5}	-1.1×10^{-5}	-1.1×10^{-5}	-1.1×10^{-5}
St Dev	0.0004	0.0004	0.0004	0.0004
Skewness	0.0724	0.0214	-0.0070	-0.0413
Kurtosis	1.8694	3.0528	4.0265	6.4370
JPN				
Average	-3.3×10^{-6}	-3.9×10^{-6}	-4.3×10^{-6}	-4.4×10^{-6}
St Dev	0.0002	0.0002	0.0002	0.0003
Skewness	0.2587	-0.0273	-0.0833	-0.1939
Kurtosis	4.5844	6.1135	7.9302	12.4972

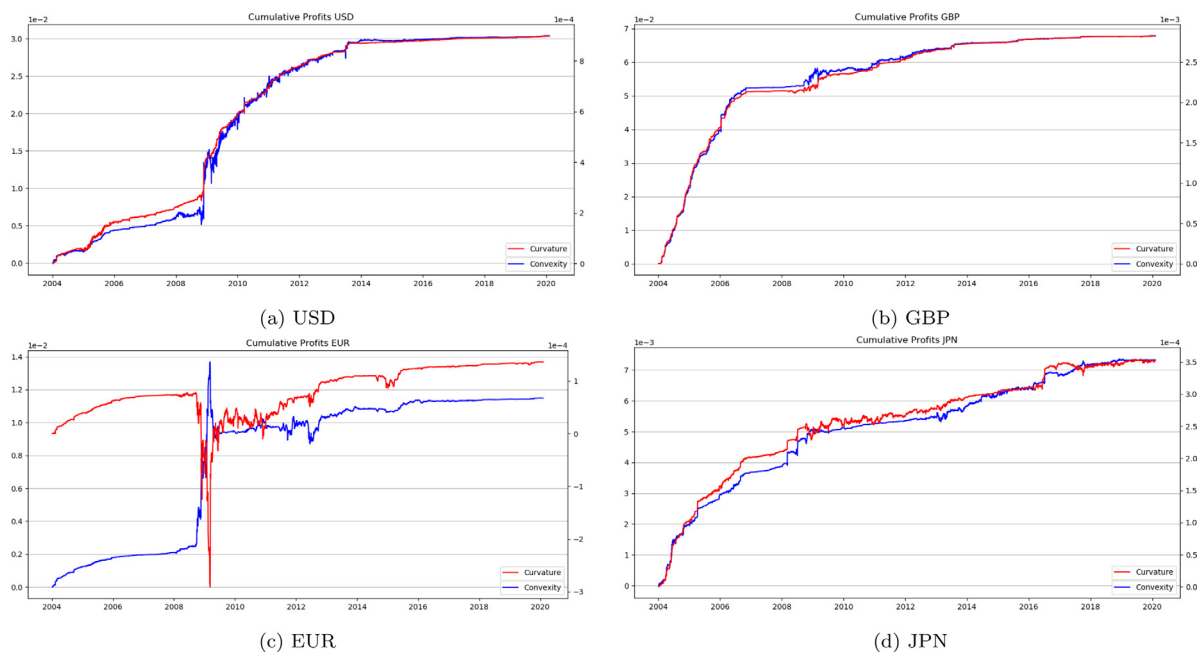


Fig. 5. In the upper row, the cumulative profit for USD (left panel) and GBP (right panel) from the curvature-based strategy (labelled ‘Curvature’) and the 2-factor convexity-based strategy (labelled ‘Convexity’). In the lower row, same for EUR (left panel) and JPN (right panel).

nothing’ about volatility) gives rise to substantial losses in this period. This should be contrasted with the convexity-based strategy, which also displays considerable volatility, but remains strongly profitable. We discuss in Section 5 why the two strategies behave so differently.

Extended periods of no profitability for both strategies indicate that in these periods the swap curve is efficiently priced. For instance, we can conclude that at least in the last five years covered by our data set in all currencies the curvature of the swap curve has been broadly consistent with the underlying volatility. However, when the strategies *are* profitable, they add to the cumulative profit either at a remarkably stable rate, or at a constantly declining rate for prolonged periods of time. This naturally raises the questions of why the profitability is localized during these prolonged distinct periods. In Section 5 we will link these periods of profitability with volatility and, above all, liquidity.

Table 2

The daily average, standard deviation, skewness and excess kurtosis for the profits made by the unconditional, curvature and convexity strategies (labelled ‘Unconditional’, ‘Curvature’ and ‘Convexity’, respectively), for the four currencies under study.

Statistic	Unconditional	Curvature	Convexity
USD			
Average	4.81×10^{-5}	2.23×10^{-7}	7.54×10^{-6}
St Dev	0.0057	0.0000	0.0002
Skewness	0.48	13.85	9.40
Kurtosis	26.51	405.79	329.85
GBP			
Average	7.07×10^{-5}	7.24×10^{-7}	1.73×10^{-5}
St Dev	0.0100	0.0000	0.0002
Skewness	0.16	8.80	8.35
Kurtosis	14.69	160.74	239.85
EUR			
Average	4.63×10^{-5}	3.82×10^{-8}	2.76×10^{-6}
St Dev	0.0030	0.0000	0.0002
Skewness	0.36	5.73	4.32
Kurtosis	22.57	213.98	147.54
JPN			
Average	-2.13×10^{-5}	9.74×10^{-8}	1.88×10^{-6}
St Dev	0.0033	0.0000	0.0000
Skewness	0.26	2.66	6.09
Kurtosis	14.96	29.66	102.91

Having discussed the qualitative features of our findings, we now present a quantitative analysis of the results. Table 2 shows the descriptive statistics for the returns for the Unconditional, Curvature and Convexity strategies.

The most noteworthy feature is the very substantial increase in right skewness and kurtosis for the distributions of the profits from the curvature- and convexity-based strategies with respect to the distribution of profits from the unconditional strategy (and, for that matter, of the rate changes displayed in Table 1). These increases in the higher moments of the curvature- or convexity-based return distributions are therefore a direct consequence of the dynamic trading strategy induced by the curvature and convexity signals.

For both strategies, the increase in kurtosis with respect to the unconditional strategy is due to the large profits or losses made by a long- or short-convexity position, respectively.²³

The positive skewness of the convexity return distribution then suggests that the convexity strategy is *long* convexity most of the times. We find that this is indeed the case for USD, GBP and EUR, where the convexity signal recommends a long convexity position in 99.3%, 94.8% and 91.2% of the times, respectively. (We note that in the case of JPN, however, the convexity signal recommends a long convexity position only 29.1% of the times.)

These different asymmetries in the return distributions therefore imply that, according to our convexity signal, the USD, GBP and EUR swap curves are on average not ‘curved enough’ (with not ‘enough’ negative carry), while the opposite is true in the case of the JPN. This is confirmed by the observation that the unconditional (long-always) strategies have positive returns in USD, GBP and EUR, but negative returns in JPN — see the top row of each panel in Table 2. However, as we discuss immediately below, these positive returns are not sufficiently large compared to their variability to make the unconditional strategies economically attractive even for a low-risk-aversion investor. This can explain why these small average biases are not arbitrated away.

The very pronounced right-skewness and the qualitative profile for the cumulative profits displayed in Fig. 5 suggest that the Sharpe ratio is ill-suited to measure the risk/reward trade-off of the various strategies. Table 3 therefore also shows the performance measures introduced in Section 3.3. More precisely, in the rows labelled ‘Unconditional’, ‘Curvature’ and ‘Convexity’ we report the annualized Goetzmann et al.’s θ return measure; the annualized management fee, F , calculated as discussed in Section 3.3; the annualized Sharpe ratio; and the relative maximum drawdown for the following strategies²⁴:

1. the unconditional strategy based on the benchmark portfolio (‘Unconditional’);
2. the conditional strategy obtained by scaling the benchmark portfolio by the strength of the curvature signal as per Eq. (14) (‘Curvature’);
3. the conditional strategy obtained by scaling the benchmark portfolio by the strength of the convexity signal as per Eq. (13) (‘Convexity’).

In the table we also show the p values calculated by simulation as described in Section 3.3 for all the performance measures. To guide the eye, all values significantly different from zero at the 90th, 95th and 99th confidence level are also marked with one,

²³ As with a delta-hedged long-gamma position in options, a duration-immunized long-convexity position in swaps will reap large gains for big *positive* or *negative* moves in the underlying, and will record small losses in time value – negative carry – when the underlying does not move enough.

²⁴ The results reported are for an ever-expanding window length. The sensitivity of the curvature results to different window lengths are shown in Table 4.

Table 3

From the second column starting from the left: the annualized Goetzmann et al. (2007) Θ return measure, the annualized performance fee, F , the Sharpe Ratio and the maximum relative drawdown ('Max DD') for the unconditional strategy ('Unconditional'), the curvature strategy ('Curvature') and the convexity strategy ('Convexity') for USD, GBP, EUR and JPN. The symbols *, **, *** indicate a performance statistic different from 0 at the 90th, 95th and 99th confidence level. The p values, reported in the rows below the measures to which they refer, are with respect to the null hypothesis, H_0 , that the signal carries no information. In the case of the Sharpe ratios, the significance levels were also calculated using the results that apply to the asymptotic distribution; these were always found in agreement with the simulation estimate.

Strategy	Goetzmann Θ	Fee F	Sharpe ratio	Max DD
USD				
Unconditional	-0.055	-0.035	0.135	0.433
Curvature	0.267***	0.286***	1.570***	0.024
p -value curvature	0.000	0.000	0.000	–
Convexity	0.067***	0.086***	0.666***	0.105
p -value convexity	0.001	0.001	0.001	–
GBP				
Unconditional	-0.065	-0.039	0.119	0.373
Curvature	0.457***	0.482***	2.492***	0.020
p -value curvature	0.000	0.000	0.000	–
Convexity	0.311***	0.337***	1.820***	0.046
p -value convexity	0.000	0.000	0.000	–
EUR				
Unconditional	-0.025	-0.007	0.264	0.955
Curvature	-0.048	-0.031	0.141	2.717
p -value curvature	0.209	0.214	0.230	–
Convexity	0.088***	0.105***	0.766***	0.209
p -value convexity	0.002	0.002	0.002	–
JPN				
Unconditional	-0.084	-0.076	-0.056	2.167
Curvature	0.215***	0.222***	1.314***	0.072
p -value curvature	0.000	0.000	0.000	–
Convexity	0.232***	0.240***	1.383***	0.047
p -value convexity	0.000	0.000	0.000	–

two or three asterisks, respectively. The maximum drawdown is expressed as a fraction of the final profitability of the strategy. So, a maximum drawdown of, say, 0.10 means that an investor can expect to lose 10% of the final wealth over a 16-year investment period in a single localized downturn. In all cases the portfolio is made up of *three* swaps and is duration-neutralized against changes in x_1 ('level') and x_2 ('slope'). The same information is presented for USD, GBP, EUR and JPN.

In all currencies, the strategies of being *unconditionally* long convexity after duration hedging with respect to two (or, as shown in the online technical appendix, three) factors have Sharpe ratios statistically indistinguishable from zero and the associated Θ performance measure and management fees, F , are always strongly negative. This is in agreement with all the findings in the literature results reported in the introduction and shows that *on average* convexity is correctly priced.

The results are totally different when we look at the conditional strategies. Indeed, Table 3 shows that when we carry out the curvature and convexity strategies described in Section 3.2, with one single exception we find positive, and sometimes very high, Θ performance measures, management fees, F , and Sharpe ratios. The one exception out of 12 currency/strategy combinations (including the 3-factor results reported in the online technical appendix) is the curvature strategy in EUR. In this case, the Sharpe ratio is positive, although not significantly different from zero, and the management fees and theta return measures worse than those for the unconditional strategy. We discuss the case of EUR in Section 5.

We find that the Θ returns or the management fees paid by an investor to enter the curvature or convexity strategies are often extremely high: well over 30% per annum for the curvature strategies in USD and JPN, and over 40% per annum for the curvature strategy in GBP. The reasons for these very high fees, F , and Θ returns are to be found (i) in the very pronounced right skewness of the return distribution (as one would associate with a long-gamma option position — see Table 2 and the discussion in Section 2) and (ii) in the very small drawdowns displayed by these strategies (these are visually apparent from Fig. 5, are quantitatively reported in last column of Table 3, and are further discussed below). As a consequence of both these features, even a considerably risk averse investor would see such small fluctuations in her total wealth that she would be willing to pay these high fees.

We also note that, given the capital we have chosen, in no case did a daily 'margin call' (change in capital required) exceed the posted capital/collateral — therefore the attractive performance statistics reported in Table 3 were not obtained by choosing an imprudent amount of capital and establishing excessive leverage.

These results are very robust. We already mentioned that different estimates of the swap rate volatility only marginally change the performance statistics. The no-peek-ahead estimation of the reversion level for the curvature plays a more delicate role for the curvature strategy, and calls for a more careful investigation. We therefore show in Table 4 the dependence of the various performance metrics on the length of the window used in the estimation of the curvature reversion level. As it is apparent, there is limited variation in the overall results when windows of different lengths or a rolling window are used. We note that having a

Table 4

Various performance metrics as a function of the length of the window used to estimate the reversion level. Equal weights were given to the observations in each window. The rows headed 'Extending window' refers to equal weights given to an ever-expanding window. The rows headed $W = 20$, $W = 60$ and $W = 80$ refer to equal weights given on a rolling window of 20, 60 and 80 business days, respectively.

Curvature	Ingersoll Θ	Fee F	Sharpe ratio	Max DD
USD				
$W = 20$	0.396	0.414	2.142***	0.015
$W = 60$	0.363	0.381	1.995***	0.017
$W = 80$	0.356	0.373	1.960***	0.017
Extending window	0.267	0.286	1.570***	0.024
GBP				
$W = 20$	0.538	0.564	2.864***	0.008
$W = 60$	0.524	0.549	2.793***	0.006
$W = 80$	0.5183	0.544	2.769***	0.006
Extending window	0.457	0.483	2.492***	0.020
EUR				
$W = 20$	-0.088	-0.070	-0.032	10.88
$W = 60$	-0.069	-0.051	0.053	5.001
$W = 80$	-0.073	-0.055	0.034	8.502
Extending window	-0.0478	-0.031	0.141	2.712
JPN				
$W = 20$	0.440	0.447	2.314***	0.029
$W = 60$	0.379	0.386	2.037***	0.057
$W = 80$	0.365	0.371	1.972***	0.058
Extending window	0.215	0.222	1.314***	0.072

fixed-length, rather than ever-expanding, window improves somewhat the metrics for USD, EUR and JPN — the currencies where the curvature strategy 'works' —, and degrades them in the case of EUR. We note in passing that a shorter window, and, therefore, a more 'local' estimate of where the curvature should revert to, would improve all performance metrics in USD, GBP and JPN. However, this information would not have been available *ex ante* to an investor, and therefore we do not pursue this angle.

As mentioned above, the key question left unanswered is what accounts for the persistent, but suddenly varying, positive or flat slopes of the cumulative profit profiles. With little exaggeration, one can say that for all strategies the cumulative profit profiles can be approximated by juxtaposing piece-wise segments which are either flat, linear or with a declining first derivative. Why is this the case? We provide a detailed explanation in the next section: in a nutshell, we show in what follows that these 'regimes' can be convincingly explained by varying market liquidity.

5. Discussion

In this section we try to understand the qualitative behaviour for the cumulative profits displayed in Fig. 5, and in particular for the clearly identifiable 'regimes' of different, non-negative slopes (derivatives with respect to time) – slopes, that is, which remain constant, or decline smoothly, over extended periods of time.

The 'derivative' (the one-period finite-difference) of the cumulative profit is, of course, the daily profit itself. To estimate these persistent slopes of the cumulative profit function, we therefore consider the average of one-period finite differences over extended periods of time. We do so by calculating the centred moving average of the daily profits.²⁵ This has the added benefit of filtering the high-frequency noise component of the signal. Using these centred moving averages of daily profits as proxies for the almost piece-wise constant slopes, we analyse the curvature-based strategies first and the convexity-based strategies then.

5.1. When is the curvature-based strategy profitable?

The profitability of the curvature strategies shows that it is possible to time the mean-reversion properties of the swap-rate curvature to reap very attractive returns. Why is this the case? Of course, the strongly mean-reverting behaviour of the curvature *per se* does not automatically offer a pseudo-arbitrage opportunity. For instance, it is well known that volatility is mean-reverting²⁶: since the curvature is a function of the volatility, the mean-reversion in the curvature could therefore simply be inherited by the mean-reversion of the volatility and the curvature could remain efficiently priced at all times. However, if the curvature mean-reverted over time by following 'as it should' the mean-reversion behaviour of the volatility, no inefficiencies would arise and no systematic profits would be made by timing the mean reversion of the curvature. But we have shown that this is not the case. How and when, then, are these profits made? We answer this question by looking at the correlation between the timing of the

²⁵ The centred moving average obviously uses peek-ahead information. However, we make use of this forward-looking information only to explain our results, not for the strategies.

²⁶ See, e.g., Foque et al. (2000).

Table 5

The intercept (α), the slope (β) and the coefficient of determination (R^2) in the regression of the cumulative profit from the curvature and convexity strategies, $PL_s(t_i) = \alpha^k + \beta^k liq^k(t_i) + \epsilon^k(t_i)$, with $k = PC1, VIX, Noise$ and $s = curvature, convexity$, obtained using as explanatory variables the liquidity proxies given by the first principal component ('PC1') in Rebonato and Sherwin (2018), the VIX index ('VIX'), and the noise measure in Hu et al. (2012) ('Noise'). The numbers in parentheses indicate the t statistic for the regression coefficients (60-day centred moving averages). The t statistics were calculated using with Newey–West correction, with the lag chosen using the criterion in Andrews and Monohan (1992).

Liquidity measure	α	β	R^2
Curvature			
PC1	2.63E–7(25.08)	1.06E–7(18.85)	0.677
VIX	–2.56E–7(–8.05)	2.66E–8(13.98)	0.557
Noise	3.39E–8(2.43)	9.00E–8(19.50)	0.699
Convexity			
PC1	2.63E–7(22.94)	1.06E–7(17.71)	0.658
VIX	–2.56E–7(–9.31)	2.66E–8(14.15)	0.568
Noise	3.39E–8(0.98)	9.00E–8(15.39)	0.639

profitability of the strategies introduced in this paper and conditions when market efficiency may be reduced. We look at this from two complementary angles, in Sections 5.2 and 5.3.

5.2. Comparison with liquidity measures

As mentioned in the Introduction, Hu et al. (2012) argue that, when liquidity is poor, the ability of pseudo-arbitrageurs to correct deviations of US Treasury prices away from their market-efficient values is hampered. In their case, the deviations from fundamentals are estimated using the pricing errors from a popular (Nelson and Siegel, 1987) fitting model that is routinely used by market practitioners to identify 'cheap' and 'dear' bonds. Hu, Pan and Wang, however, are not interested in analysing a trading strategy, but in highlighting the link between price discrepancies (the 'noise' in their title) and reduced liquidity. It should be stressed that the link in Hu et al. (2012) between 'noise' and liquidity is somewhat indirect: they *measure* price inefficiencies and *deduce* that liquidity must have deteriorated. However, Nguyen et al. (2018) show the existence of a strong link between Hu, Pan and Wang's noise and their own direct microstructural measure of liquidity. This helps establishing a transmission mechanism from reduced liquidity to price inefficiencies, and suggests that the same mechanism can apply to *any* deformation of the swap curve away from fundamentals — in particular to deviations of the curvature away from where 'it should be'.

If this were correct, we should find a strong correspondence between the times when the US curvature strategy is most profitable and the periods of reduced liquidity. To test this hypothesis, we look at a number of liquidity estimators, including Hu, Pan and Wang's 'noise', and we use them as explanatory variables for a moving average of the profits reaped from the curvature-based strategy:

$$PL_{curv}(t_i) = \alpha^k + \beta^k liq^k(t_i) + \epsilon^k(t_i) \quad (20)$$

where $liq^k(t_i)$ is time t_i value of the k th liquidity estimator and $PL_{curv}(t_i)$ is the (centred moving average of the) profit reaped by the curvature strategy in USD from time t_i to time t_{i+1} . The three liquidity measures considered were the Hu, Pan and Wang's 'noise as liquidity' (NOISE); the first principal component of a number of liquidity indicators (PC1), as presented in Rebonato and Sherwin (2018); and the VIX index, which has been routinely employed as a measure of market distress (see, e.g., Whaley (2000)).

When we estimate Eq. (20) we find very high coefficients of determination, with R^2 of 0.677, 0.557 and 0.699 for the PC1, VIX and NOISE liquidity measures, respectively, and t -statistics for the β coefficients of 18.85, 13.98 and 19.50, respectively.²⁷ The full results of the regressions are shown in Table 5 for the case of a 60-day centred moving average of daily returns.²⁸ The time series of the profits from the curvature strategy and of the various liquidity proxies are then shown in the left-hand panels of Fig. 6. (The right-hand panels are discussed in the following section.)

We also note that Rebonato and Sherwin (2018) analyse in detail the mean-reverting behaviour of liquidity, and identify liquidity proxies with high and low reversion speeds. In particular they estimate reversion speeds of 0.53 ± 0.30 , 0.57 ± 0.30 and 0.57 ± 0.31 for the On/Off spread, the emerging market spread and the high-yield spread, respectively, which constitute their set of slowly-mean-reverting liquidity proxies.²⁹ These reversion speeds are important, because they are linked to the characteristic time over which liquidity is restored to the market after a shock. After filtering periods shorter than two weeks, we find for the reversion speed of the USD curvature a value of 0.534 ± 0.12 , which is closely aligned with the reversion speeds of the slow-reverting proxies in Rebonato and Sherwin.

Taken together, these results indicate that periods of profitability are strongly associated with periods of poor liquidity, and that the *decay* in profitability that we observe in the segments of the cumulative-profit profiles with decreasing first derivative occurs at

²⁷ The t statistics were calculated using the Newey–West correction. We have calculated the lag selection parameter for the standard Newey–West HAC estimate (as in Andrews and Monohan (1992)) as $\max Lag = \text{floor}(4 * (T/100)^{2/9})$. The standard Newey–West OLS coefficient covariance was then estimated by setting the bandwidth to $\max Lag + 1$.

²⁸ We found that the results remain qualitatively similar when we change the length of the window of the moving average between 20 and 80 business days.

²⁹ Their fast-reverting proxies have reversion speeds ranging from 1.15 to 3.91.

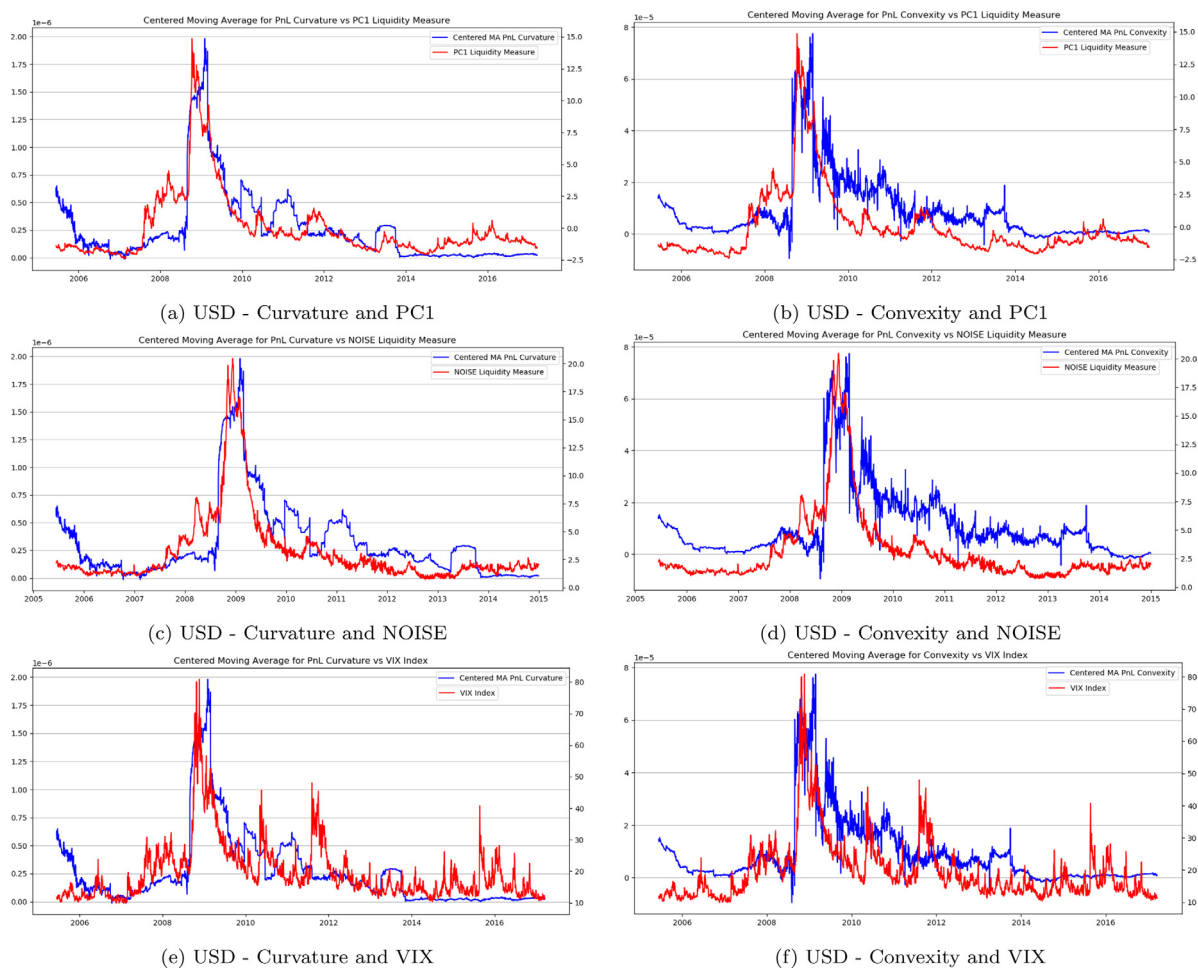


Fig. 6. In the first row, the time series of the profits made by the curvature strategy (left panel) and convexity strategy (right panel) plotted with the [Rebonato and Sherwin \(2018\)](#) PC1 liquidity measure. In the second row, the time series of the profits made by the curvature strategy (left panel) and convexity strategy (right panel) plotted with the NOISE liquidity measure. In the third row, the time series of the profits made by the curvature strategy (left panel) and convexity strategy (right panel) plotted with the VIX liquidity measure.

a rate consistent with the recovery in liquidity. This suggests a strong link between the periods of high profitability for the curvature strategy and periods of reduced liquidity. The same financial mechanism proposed by Hu et al. to explain the existence of exploitable inefficiencies in the *level* of US Treasuries therefore accounts convincingly for the profitability of the *curvature* strategies discussed in this work. Similar considerations can be drawn for our convexity trades, but the discussion requires some qualifications, that we present in Section 5.4. Before that, we explore the links between the timing of our convexity trades and of the [Cieslak and Povala \(2015\)](#) return-predicting factors.

5.3. Comparison with the Cieslak-Povala cycles

As mentioned in the Introduction, [Cieslak and Povala \(2015\)](#) explore the profitability of trades in the US Treasury market based on buying a long-maturity bond and funding the purchase with a one-period bond. Their return-predicting factor is a combination of cycles, which are in turn built as the residuals of the regression of yield levels against a slow-moving average of inflation. These cycles have a strong predictive power for excess returns and the authors explain the profitability of the return-predicting factor as due to temporary deviations of prices from fundamentals (from where yield levels ‘should be’ given inflation). The same explanation is offered in [Radwanski \(2002\)](#) and [Rebonato and Ronzani \(2020\)](#).

We stress that the strategy in Cieslak and Povala is designed to exploit deviations of the *level* of the yield curve, and refers to Treasury bonds, not swaps. Nonetheless, in this section we want to test further the hypothesis that conditions of impaired liquidity are common across US fixed-income markets and therefore also affect the strength of the signals for the convexity and curvature

Table 6

The intercepts, (α), slope (β), and the coefficient of determination (R^2) in the regression, $signal_t^x = \alpha^x + \beta^x Cycle_t^i + \epsilon_t^x$, of the convexity and curvature signals (index x) obtained using different Cieslak–Povala cycles, $Cycle_t^i$, as explanatory variables. Cycle 1 is associated with the shortest maturity (1 year) and Cycle 5 with the longest maturity (9 years). The numbers in parentheses indicate the t statistic for the regression coefficients. Note that all the α and β coefficients are significantly different from 0 at the 99% confidence level. The t statistics were calculated using Newey–West correction, with the lag chosen using the criterion in Andrews and Monohan (1992).

Signal of the strategy	α	β	R^2
Cycle 1			
Signal convexity	0.0108(8.538)	−0.758(−6.779)	0.549
Signal curvature	0.0002(6.463)	0.0063(4.193)	0.311
Cycle 2			
Signal convexity	0.0095(8.606)	−0.877(−7.119)	0.545
Signal curvature	0.0002(6.330)	0.0071(3.899)	0.292
Cycle 3			
Signal convexity	0.0092(8.585)	−0.955(−6.387)	0.447
Signal curvature	0.0002(5.875)	0.0076(3.515)	0.235
Cycle 4			
Signal convexity	0.0100(6.156)	−0.914(−5.419)	0.302
Signal curvature	0.0002(5.356)	0.0078(3.128)	0.181
Cycle 5			
Signal convexity	0.0115(5.304)	−0.767(−4.324)	0.173
Signal curvature	0.0002(4.864)	0.0075(2.736)	0.137

trades.³⁰ If this were the case, then the timing of the Cieslak and Povala signal would be similar to the timing of the convexity and curvature signals. To check whether this is the case we therefore regress the curvature and convexity signals against the five Cieslak and Povala cycles³¹:

$$sign_t^x = \alpha_i^x + \beta_i^x Cycle_t^i + \epsilon_t^i \quad (21)$$

where $x = curv, conv$ denotes either the curvature or the convexity signal, and $Cycle_t^i$ signifies the i th Cieslak and Povala cycle ($i = 1, 2, \dots, 5$).

The results of the regression are presented in Table 6, and they show a strong dependence of the strength of the curvature and convexity signals and the Cieslak and Povala cycles: all the slopes are strongly statistically significant (all well above the 99th confidence level), and with coefficients of determination ranging from 0.549 to 0.137.³² We note in passing that for all the cycles the slope coefficients are negative for the convexity signal and positive for the curvature signal. This is a consequence of the fact that the curvature signal is by construction positive and negative — it is a ‘cycle’ itself —, while the convexity signal is on average positive.

In sum: we find a statistically very strong relationship between periods when the Cieslak and Povala cycles deviate from their reversion levels and when the convexity and curvature signals are strong. Taken together with the liquidity findings in Section 5.2, both sets of results strongly substantiate the intuition that the mean reversion of the curvature is exploitable because in periods of poor liquidity the swap curvature can deviate from where it should be, and that these deviations are not promptly corrected because they occur in periods when arbitrage capital is not forthcoming. In the next section we analyse in more detail the differences in the timing of the profitability of the curvature and convexity strategies.

5.4. What is the difference between the curvature- and convexity-based strategies?

The discussion and results presented in the previous section nicely frame the question of what distinguishes the convexity and curvature strategies. We have shown that the mean-reverting behaviour of the curvature can to a large extent be explained by the similarly mean-reverting behaviour of liquidity. However, the curvature strategy makes no explicit use of volatility information, which is instead a key component of the convexity strategy. Does the extra volatility information improve the performance of the convexity strategies over the curvature ones?

First, we confirm that also the profitability of the convexity strategies is very strongly linked to the liquidity measures discussed before. The right-hand panel of Fig. 6 shows that this is indeed the case. The R^2 coefficients of the regressions

$$PL_{conv}(t_i) = \alpha^k + \beta^k liq^k(t_i) + \epsilon^k(t_i), \quad k = Noise, PC1, VIX \quad (22)$$

³⁰ In the case of the Cieslak and Povala cycles we compare signals rather than profits, because it does not make sense to compare the *yearly* returns with monthly overlap of the Cieslak and Povala study with the *daily* returns from our analysis. Since we only have the Cieslak and Povala data at monthly intervals, we use in the regression in Eq. (21) the monthly average of the curvature signal.

³¹ Cycle 1 is associated with the shortest maturity (1 year) and Cycle 5 with the longest maturity (9 years).

³² Also in this case the t statistics were estimated using the Newey–West correction, with the lag chosen using the criterion in Andrews and Monohan (1992).

are 0.658 for PC1, 0.568 for VIX and 0.639 for NOISE,³³ with t -statistics for the β coefficient of 17.71, 14.15 and 15.39, respectively.³⁴ This confirms that, at least in the case of USD for which we have reliable liquidity measures, also the convexity profits are associated with periods of poor liquidity, when inefficiencies can arise and persist.

Looking at the performance metrics presented in Table 3, were it not for the case of EUR one may be tempted to conclude that the curvature strategy, which is simpler to implement and, by requiring the estimation of fewer quantities, presumably more robust, is more attractive than the convexity strategies. Its shortcomings are however vividly shown in the days when the EUR swap curve moved as dramatically as it happened on December 2 and 3, 2008. (See the discussion in Section 4.1.) Over this period the EUR curvature becomes very pronounced. If one takes the deviation of the curvature from its reversion level as a trading signal, this extremely negative curvature suggests to be short convexity in expectation of a reversion to the mean. This gives rise to a short-convexity ('short-gamma') position. Given the wild moves experienced in this period, this volatility-unaware strategy proves disastrous. The convexity signal, on the other hand, can make use of the extra information from the volatility and 'knows' that, despite the very strong negative carry, the expected profit from re-hedging will more than compensate for the 'option premium' paid. Therefore, it correctly suggests a *long-convexity* position, and it manages to do so despite our volatility estimate being backward-looking.

Apart from this striking 'test case', the same message is conveyed by a quantitative analysis of the signals and returns associated with the various strategies. The reasoning goes as follows. First, we note that the correlations between the convexity and the curvature signals are all significantly *negative*: -27.6% for GBP, -22.4% for USD, -21.9% for JPN and -39.8% for EUR for the two-factor, three-bond case. What is particularly relevant for the argument to follow is that these negative correlations are largely due to the curvature and convexity signals giving different-sign indications in periods of high volatility. Indeed, when we sort the curvature and convexity signals by the magnitude of the average swap rate volatility, we find that on the 100 most volatile days the curvature and convexity signal have opposite sign 100 times in EUR, 98 times in GBP, 98 times in JPN and 76 in USD.

Next, we find a strong positive correlation between volatility and swap curve curvature. Therefore the swap curve tends to be more negatively curved in periods of high volatility. As the negative correlations between signals reported above show, in these situations of market turmoil the recommendations of the two signals are often different. More precisely, in these periods of high volatility and high curvature the suggestion from the convexity signal tends to be *long* convexity, and from the curvature signal to be *short* convexity. As we have seen, this can give rise to very large losses for the latter strategy.

We stress that this analysis only captures the gross features of the co-dependence between the curvature and convexity signals in the four currencies. In the online technical appendix we show and briefly discuss the copula densities for the four couples of signals, which clearly display very complex (and currency-specific) tail-dependencies of the two signals. The key message however remains valid: in normal market conditions the curvature and convexity signals are highly correlated, and therefore the cumulative profits grow at a very similar rate (see Fig. 5); in excited market conditions the positive correlation between the curvature and convexity signals breaks down, and complex patterns of co-dependence arise, which de-synchronize the rate of growth of the cumulative profits. And in these turbulent market conditions, the volatility-aware convexity signal tends to 'get it right' more often than the curvature signal.

On the basis of these considerations we can therefore draw the following conclusions: in normal conditions the curvature strategy may well be more robust, and therefore perform somewhat better. However, since it lacks any information about the trade-offs between curvature and volatility, it can produce totally inappropriate signals in periods of high distress. Given the strongly asymmetric nature of volatility shocks and the dangers of being short convexity during large moves, the extra information conveyed by the convexity signal therefore provides a valuable degree of insurance.

5.5. Transaction costs

In this part of the analysis we consider the effect of transaction costs on the profitability of the strategies presented above.

From discussions with trading desks of a large asset manager, the bid/offer spread required by a plain-vanilla trader will typically be a fraction of a basis point (hundredth of a percentage point) times the *net* duration of the position. At the start of each transaction period the portfolio of swaps is always immunized against moves in 'level' and 'slope' – our x_t^1 and x_t^2 factors (and, for the results presented in the on-line technical appendix against changes in 'curvature', our x_t^3 factor, as well). Therefore the initial bid/offer charge will be small, and we have very conservatively set it at 1 basis point of the notional of the longest-dated swap. After each time step, the portfolio will no longer be duration immunized and a rebalancing transaction will have to be executed with the plain-vanilla desk. We have conservatively assumed that the hedge provider will require one and a half basis point times the absolute value of the net duration of the rebalancing portfolio of swaps. In order to account for a widening bid-offer spread in periods of high volatility, we have also doubled the charge to three basis points if the volatility of the underlying swaps is in the upper decile of the volatility distribution.

When these charges are included, the resulting effect on the performance measure Θ and on the management fee, F , remains modest – of the order of a tens of basis points per annum at most, to be compared with positive Θ and management fees, F , of many percentage points. The changes in Sharpe ratios were similarly small. For the sake of brevity, we therefore do not report the new performance statistics, but these as well are available in the on-line technical appendix.³⁵

We can therefore conclude that including the effect of transaction costs does not change any of the qualitative conclusions drawn in the analysis so far and barely alters the quantitative results.

³³ These results were obtained for a 60-day moving average. Once again, we found that the results remain qualitatively similar when we change the length of the window of the moving average between 20 and 80 business days.

³⁴ Also in this case the t statistics were estimated using the Newey–West correction, with the lag chosen using the criterion in Andrews and Monohan (1992).

³⁵ See worksheet 'Transaction Costs', and Table TC2 in particular.

6. Conclusions and suggestions for further studies

Looking at the problem of convexity through an option-theoretical lens, we have provided in this study a precise criterion to establish whether at any given point in time the long-end curvature of four major swap markets is correctly priced. We have shown that, under mild assumptions, the correct curvature is a function of the steepness of the yield curve, of the net ‘carry’ and of the volatility of the underlying swap rates. By employing this tool we have first confirmed that, *on average*, the USD, GBP, EUR and JPN swap curves are close to being efficiently priced: USD, GBP and EUR seem to display a bit too little, and JPN a bit too much, curvature, but the average deviations from efficiency are too small to be economically exploitable.

We have however shown that, by employing convexity- and curvature-based no-peek-ahead strategies, it is possible to ‘time convexity’, i.e., to generate economically attractive abnormal returns with no-peek-ahead and transaction-cost-aware dynamic trading strategies. The fees a risk-averse investors would be willing to pay a manager who engaged in these strategies are high, and robust across different strategy implementations and currencies.

In the spirit of the study by [Hu et al. \(2012\)](#), we have explained why these abnormal returns arise by establishing a strong correlation between the profits made by the strategies and several measures of liquidity. These conclusions are corroborated by the analysis we carry out of the timing of the curvature/convexity signal of the one hand, and of the [Cieslak and Povala \(2015\)](#) cycles on the other. The picture that emerges is that, in period of high volatility and poor market liquidity, arbitrage capital is not forthcoming and deviations of the curvature from its equilibrium value are not traded away. When conditions of good liquidity prevail again the deviations are corrected, and the curvature becomes efficiently priced again.

The approach we have taken naturally points to several and interesting related lines of investigation. The ‘long-gamma’ profile in [Fig. 2](#), for instance, raises a few obvious questions: how much would it cost to reproduce this profile using a straddle of long-dated swaptions? Would the implicit premium from the negative carry be greater or smaller than the option premium? In other words, at any point in time would it be more efficient to ‘buy or sell convexity’ via swaptions, or by exploiting the curvature of the swap curve? Could one create a delta-, gamma- and vega-neutral portfolio of long-dated swaps and swaption straddles? And would this package have a non-zero cost? We would love to carry out these investigations, but we do not have reliable swaption prices at our disposal, and we therefore leave these topics for future analysis.

Appendix A. Derivation of the duration-neutralizing weights

Changes in the variables x_t^i are given by:

$$\Delta x_t^1 = a_{11}\Delta S R_t^1 + a_{12}\Delta S R_t^2 + a_{13}\Delta S R_t^3 \quad (23)$$

$$\Delta x_t^2 = a_{21}\Delta S R_t^1 + a_{22}\Delta S R_t^2 + a_{23}\Delta S R_t^3 \quad (24)$$

$$\Delta x_t^3 = a_{31}\Delta S R_t^1 + a_{32}\Delta S R_t^2 + a_{33}\Delta S R_t^3 \quad (25)$$

where $S R_t^i$ denotes again the time- t value of the equilibrium swap rate for the i th swap. In matrix form

$$\Delta x = A \Delta S R \quad (26)$$

and

$$\Delta S R = A^{-1} \Delta x \equiv B \Delta x \quad (27)$$

with $B = A^{-1}$. Consider then a portfolio, Π_t made up as follows:

$$\Pi_t = w_1 S_t^1 + w_2 S_t^2 + S_t^3 \quad (28)$$

We want to immunize the portfolio against changes in some or all of the new variables, x . In order to duration hedge against x_1 (‘level’) and x_2 (‘slope’) we need three swaps and four if we also want to hedge against x_3 , (‘curvature’). The matrix A for three or four swaps was chosen to be

$$A = \begin{pmatrix} 1 & 1 & 1 \\ -1 & 0 & 1 \\ 1 & -2 & 1 \end{pmatrix} \quad (29)$$

and

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & -0.36 & 0.36 & 1 \\ 0.8 & -0.8 & -0.8 & 0.8 \\ 0.25 & -0.6944 & 0.6944 & -0.25 \end{pmatrix} \quad (30)$$

The rationale for this choice is explained in [Section 3](#). We focus in what follows in immunization against level and slope, as the formulae are simpler and the extension to four factors obvious.

We want to find weights, w_1 and w_2 such that the portfolio is first-order immunized (i) against changes in x_t^1 and x_t^2 :

$$\frac{\partial \Pi_t}{\partial x_t^i} = 0 \quad (31)$$

We have

$$\frac{\partial \Pi_t}{\partial x_t^j} = \sum_k \frac{\partial \Pi_t}{\partial S_t^k} \frac{\partial S_t^k}{\partial SR_t^k} \frac{\partial SR_t^k}{\partial x_t^j} \quad (32)$$

The various term are as follows:

$$\frac{\partial \Pi_t}{\partial S_t^k} = w_k \quad (33)$$

for $k = 1, 2$ (with $w_3 = 1$).

Then

$$\frac{\partial S_t^k}{\partial SR_t^k} = Ann_t^k \quad (34)$$

and

$$\frac{\partial SR_t^k}{\partial x_t^j} = b_{kj} \quad (35)$$

Therefore if one wants to hedge against changes in x_t^1 and x_t^2 one can write

$$Cw = y \quad (36)$$

with

$$C = \begin{pmatrix} Ann_t^1 b_{11} & Ann_t^2 b_{21} \\ Ann_t^1 b_{12} & Ann_t^2 b_{22} \end{pmatrix} \quad (37)$$

$$w = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \quad (38)$$

and

$$y = \begin{pmatrix} -Ann_t^3 b_{31} \\ -Ann_t^3 b_{32} \end{pmatrix} \quad (39)$$

This gives for the weights

$$w = C^{-1}y \quad (40)$$

A similar expression applies (with the matrices now 3×3 and 3×1), if one wanted to hedge against changes in x_t^1 , x_t^2 and x_t^3 .

Appendix B. Derivation of Eq. (9)

For a receiver swaption the various terms in Ito's lemma are as follows³⁶:

$$\frac{\partial S_t^T}{\partial t} = K - f_t^r \quad (41)$$

$$\frac{\partial S_t^T}{\partial SR_t^T} = -Ann_t^T + [K - SR_t^T] \frac{\partial Ann_t^T}{\partial SR_t^T} \quad (42)$$

and

$$\frac{\partial^2 S_t^T}{\partial d S_t^{T^2}} = -\frac{\partial Ann_t^T}{\partial SR_t^T} - \frac{\partial Ann_t^T}{\partial SR_t^T} + [K - SR_t^T] \frac{\partial^2 Ann_t^T}{\partial S R_t^{T^2}} \quad (43)$$

For any at-the-money (equilibrium) swap, the first and second derivatives with respect to the swap rate simplify to

$$\frac{\partial S_t^T}{\partial SR_t^T} = -Ann_t^T (SR_t^T) \quad (44)$$

and

$$\frac{\partial^2 S_t^T}{\partial d S_t^{T^2}} = -2 \frac{\partial Ann_t^T}{\partial SR_t^T} \quad (45)$$

³⁶ To obtain this term we make the assumption that the swap accrues on a continuous basis, rather than on a discrete daily basis.

Putting the pieces together, we have

$$dS_t^T = [K - f_t^r]dt - Ann_t^T(SR_t^T)dSR_t^T - \frac{\partial Ann_t^T}{\partial SR_t^T}\sigma_{SR}^2 dt \quad (46)$$

Appendix C. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jempfin.2021.07.011>.

References

- Ahmed, N., Natarajan, T., Rao, K., 1974. Discrete cosine transform. *IEEE Trans. Comput.* C-23, 90–93.
- Andrews, D., Monahan, J., 1992. Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica* 59, 817–858.
- Ben Dor, A., Chen, Y., Dynkin, L., Hyman, J., 2006. Can Convexity be Exploited? Lehman Brothers Working Paper, Fixed Income Research.
- BIS, 2018. OTC Derivatives Statistics at End-June 2018. Bank for International Settlements, Statistical Release.
- BIS, 2019. Minimum Capital Requirement for Market Risk. Basel Committee on Banking Supervision.
- Black, F., Karasinski, P., 1991. Bond and option pricing when short rates are lognormal. *Financ. Anal. J.* 47, 52–59.
- Brown, R., Schaefer, S., 2000. Why Long Term Forward Interest Rates (Almost) Always Slope Downwards. London Business School Working Paper.
- Cieslak, A., Povala, P., 2015. Expected returns in treasury bonds. *Rev. Financ. Stud.* 10, 2859–2901.
- Cieslak, A., Povala, P., 2016. Information in the term structure of yield curve volatility. *J. Finance* LXXI (3), 1393–1435.
- Cox, J., Ingersoll, J., Ross, S.A., 1985. A theory of the term structure of interest rates. *J. Econometrics* 53, 385–407.
- Crump, R.K., Gospodinov, N., 2019. Deconstructing the Yield Curve, Vol. 884. Federal Reserve Bank of New York Staff Report.
- Dai, Q., Singleton, K., Yang, W., 2004. Predictability of Bond Risk Premia and Affine Term Structure Models. Working Paper, Stern School of Business.
- Diebold, F., Rudebusch, G., 2013. *Yield Curve Modelling and Forecasting – The Dynamic Nelson-Siegel Approach*. Princeton University Press, Princeton, NJ, and Oxford, UK.
- Elliot, G., Timmermann, A., 2016. *Economic Forecasting*. Princeton University Press, Princeton, NJ, and Oxford.
- Fabozzi, F., 1999. *Duration, Convexity and Other Bond Risk Measures*. John Wiley, Chichester.
- Foque, J., Papanicolaou, G., Sircar, K., 2000. Mean-reverting stochastic volatility. *Int. J. Theor. Appl. Finance* 3 (1), 101–142.
- Goetzmann, W., Ingersoll, J., Spiegel, M., Welch, I., 2007. Portfolio performance manipulation and manipulation-proof performance measures. *Rev. Financ. Stud.* 20, 1503–1546.
- Gupta, A., Subrahmayam, M., 1998. An empirical examination of the convexity bias in the pricing of interest rate swaps. *J. Financ. Econ.* 55 (2), 239–279.
- Gurkaynak, R., Sack, B., Wright, J., 2007. The US treasury yield curve: 1961 to the present. *J. Monetary Econ.* 54, 2291–2304.
- Henrad, M., 2014. *Interest Rate Modelling in the Multi-Curve Framework*. Palgrave MacMillan.
- Hu, G., Pan, J., Wang, J., 2012. Noise as information for illiquidity. *J. Finance* 68 (6), 2341–2382.
- Hull, J., White, A., 1990. Pricing interest rate derivative securities. *Rev. Financ. Stud.* 3, 573–592.
- Kang, J., Chen, A., 2002. Evidence on theta and convexity in treasury returns. *J. Fixed Income* 12 (1).
- Kessler, S., Scherer, B., 2009. Varying risk premia in international bond markets. *J. Bank. Financ.* 33 (8), 1361–1375.
- Khan, R., Lochoff, R., 1990. Convexity and exceptional returns. *J. Portf. Manage.* 16 (2).
- Litterman, R., Scheinkman, J., 1991. Common factors affecting bond returns. *J. Fixed Income* 1 (1), 54–61.
- Moreira, A., Muir, T., 2017. Volatility-managed portfolios. *J. Finance* LXXII (4), 1611–1644.
- Nelson, C., Siegel, A., 1987. Parsimonious modeling of yield curves. *J. Bus.* 60, 473–489.
- Nguyen, G., Engle, R., Fleming, M., Ghysels, E., 2018. Liquidity and Volatility in the U.S. Treasury Market. Federal Reserve Bank of New York, Staff Report.
- Radwanski, J., 2002. Understanding Bond Risk Premia Uncovered by the Term Structure. Working Paper, Vienna Graduate School of Finance, Vienna University of Economics, (VGSF).
- Rebonato, R., 2002. *Modern Pricing of Interest-Rate Derivatives*. Princeton University Press, Princeton, NJ, and Oxford, UK.
- Rebonato, R., Putiatyn, V., 2018. The value of convexity: A theoretical and empirical investigation. *Quant. Finance* 18 (1), 11–30.
- Rebonato, R., Ronzani, R., 2020. Predicting future yields and risk premia: The ‘blue-dots’ affine model. *J. Fixed Income*.
- Rebonato, R., Sherwin, H., 2018. Robust and Interpretable Liquidity Proxies for Market and Funding Liquidity. EDHEC Working Paper.
- Schoenmakers, J., Coffey, B., 2003. Systematic generation of parametric correlation structures for the LIBOR market model. *Int. J. Theor. Appl. Finance* 5 (5).
- Shyy, G., Lieu, C., 1994. A Note on Convexity and Bond Performance. Financial Management Association.
- Strang, G., 2003. The discrete cosine transform. *SIAM Rev.* 41, 135–147.
- Thornton, D.L., Valente, G., 2012. Out-of-sample predictions of bond excess returns and forward rates: An asset allocation perspective. *Rev. Financ. Stud.* 25 (10), 3141–3168.
- Vasicek, O., 1977. An equilibrium characterization of the term structure. *J. Financ. Econ.* 5, 177–188.
- Whaley, R., 2000. The investor fear gauge. *J. Portf. Manage.* 26 (3), 12–17.